Computers in Industry 180 (2026) 104501



Contents lists available at ScienceDirect

Computers in Industry

journal homepage: www.sciencedirect.com/journal/computers-in-industry





Digital twin-driven pose estimation and trajectory prediction for modular integrated construction on-site stacking via spatio-temporal feature fusion modeling☆

Yujie Han [a,b], Kexin Sun [a,b], Zhiheng Zhao [a,b,*], Yantao Yu [c], George Q. Huang [a,b]

[a] Department of Industrial and Systems Engineering, The Hong Kong Polytechnic University, Hong Kong, China
[b] Research Institute for Advanced Manufacturing, The Hong Kong Polytechnic University, Hong Kong, China
[c] Department of Civil and Environmental Engineering, The Hong Kong University of Science and Technology, Hong Kong, China

ARTICLE INFO

Keywords:
Modular integrated construction (MiC)
Digital twin (DT)
Pose estimation
Trajectory prediction
Spatio-temporal feature fusion

ABSTRACT

High stacking precision is essential to prevent structural collisions, reduce safety accidents, and enhance project efficiency in modular integrated construction. However, current stacking precision is constrained by delayed hoisting control, limited perception, and fragmented information sharing. Therefore, this paper proposes a digital twin-driven pose estimation and trajectory prediction system via spatio-temporal feature fusion modeling, leveraging real-time information from high-fidelity digital twins. First, physical entities are transformed into smart stacking objects that combine static inherent properties with real-time dynamic activities captured by Ultra-Wideband sensors and Inertial Measurement Units. Then, a digital twin of stacking operations is used to interact with their physical counterparts in real time, enabling seamless mapping and visualization of the stacking process. Third, the PosTraFormer network is proposed for 6-dimensional module pose estimation and trajectory prediction based on spatio-temporal fusion modeling of trajectory point relations, to provide stacking guidance and improve installation precision. Both visualization and quantitative analysis from comparative experiments and case studies demonstrate the significant advantages of the proposed approach.

1. Introduction

As a disruptive innovation, Modular Integrated Construction (MiC) is increasingly replacing traditional methods, offering benefits such as improved quality, simplified management, and shorter project durations (Pan and Hon, 2020; Zheng et al., 2020). During the MiC construction process, efficient module stacking is critical for timely assembly, high construction quality, and safety (Shan et al., 2019; Zhai et al., 2019). In practice, module stacking still depends on manual observation and the expertise of crane operators (Zhu et al., 2022), which gives rise to several critical challenges. First, crane operators often exercise delayed control due to insufficient real-time information for accurate perception of module status, thereby increasing the likelihood of safety incidents, stacking errors, and module collisions (Kim and Chi, 2021). Second, the lack of timely and integrated information sharing among stakeholders causes delays and fragmentation, which hinder coordination and perception across the project team (Zhao et al., 2022). These limitations further result in module accumulation in buffer zones, thereby disrupting upstream production and timely delivery (Yue Zhai and Huang, 2017; Zhong et al., 2017). Thus, these challenges raise higher requirements to facilitate module stacking precision, including but not limited to: (1) comprehensive module stacking perception to replace workers' observation; (2) efficient information transmission and sharing among stakeholders to avoid information silos; (3) predictive stacking guidance to refrain from delayed control of crane operators (Yue Zhai and Huang, 2017; Salama et al., 2017).

Collectively, these requirements highlight a central need: enabling stakeholders to both perceive ongoing conditions and anticipate forthcoming events. Therefore, global perceptual monitoring supported by real-time simulation of the stacking process and trajectory prediction is required. Precise simulation of the module stacking process demands real-time data collection and processing, highly accurate dynamic modeling, and extensive information sharing and collaboration. Traditional

☆ The authors would like to express their sincere thanks to the financial support from the National Natural Science Foundation of China (No. 52305557), Hong Kong Research Grants Council (No. 15203025, T32-707/22-N, C7076-22GF, C6044-23GF, R7036-22), Guangdong Basic and Applied Basic Research Foundation (No. 2024A1515011930), Research Institute for Advanced Manufacturing (RIAM) of The Hong Kong Polytechnic University (No. CDLU, CDLM, CDJX).

* Corresponding author at: Department of Industrial and Systems Engineering, The Hong Kong Polytechnic University, Hong Kong, China.
E-mail addresses: yujiejenny.han@connect.polyu.hk (Y. Han), zhiheng.zhao@polyu.edu.hk (Z. Zhao).

https://doi.org/10.1016/j.compind.2026.104501
Received 10 September 2024; Received in revised form 30 September 2025; Accepted 20 May 2026
Available online 9 June 2026
0166-3615/© 2026 Elsevier B.V. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

simulation techniques often fail to dynamically adapt to changes in the physical environment and lack the required level of interconnectivity and coordination for complex tasks such as module stacking. In contrast, Digital Twin (DT) technology provides a robust solution by replicating physical processes in a virtual environment and integrating sensors for real-time monitoring and prediction (Zhou et al., 2019; Zhao et al., 2022; Liu et al., 2021). Within the DT framework, module stacking is redefined to include real-time visibility, cyber–physical connectivity, and traceability. Furthermore, a DT is incorporated into collaborative platforms that facilitate access to and sharing of module information among stakeholders, enhancing information synchronization. Consequently, this paper aims to leverage the advantages of DT to achieve comprehensive visibility and traceability at module stacking sites through real-time data collection and positioning.

Nevertheless, addressing delayed control and achieving predictive guidance requires further development of precise methods for module pose estimation and trajectory prediction within the DT framework. Computer vision-based methods have been proposed by researchers and practitioners (Zhang et al., 2019). Yang et al. (2023) suggested surveillance cameras for environmental perception and worker trajectory prediction to boost construction safety. Similarly, Cheng et al. (2022) introduced a stereo-vision technique for aligning precast concrete bridge components. Considering that vision-based methods often fail to capture accurate 3D spatial relationships, Xie et al. (2025) proposed a framework that leverages point cloud data to detect human–object interactions on construction sites. However, on-site practices reveal that these methods struggle in dynamic and variable construction environments, poorly addressing challenges like cluttered backgrounds, wind, vibrations, fluctuating lighting, and limited camera angles (Kopsida and Brilakis, 2020; Omar and Nehdi, 2016; Liu et al., 2023). In contrast, Internet of Things (IoT) sensors can capture high precision, multisource stacking data in real-time, offering superior environmental adaptability and reliability. Combined with IoT devices, high fidelity virtual models in the DT framework can reflect the dynamic state, behavior, and changes of the physical stacking system in a comprehensive perspective (Zhou et al., 2019; Zhao et al., 2022; Liu et al., 2021).

Current methods for predicting trajectories with IoT sensors primarily use temporal models to analyze historical motion data sequences (Wang et al., 2022; Yang et al., 2023; Zhang and Ge, 2022). These sequential relationships expose the temporal continuity and directional trends of motions, enabling more reliable future trajectory predictions. However, in complex spatial movements like module rotation or flipping, reliance solely on temporal relationships for the module trajectory prediction proves challenging. This is mainly because such movements entail multi-dimensional displacement and spatial pose adjustments. This difficulty often requires deep learning networks with advanced capabilities for handling complex nonlinear problems. Recently, deep learning models have achieved outstanding performance in multiple domains (Shang et al., 2023; Han et al., 2024; Gao et al., 2025). Specifically, in tasks that involve extracting complex spatiotemporal features, Zhang et al. (2024) designed a multi-attention network based on spatio-temporal feature fusion to predict pedestrian crossing intention. Ali et al. (2025) proposed a spatio-temporal graph convolutional recurrent network with multi-scale feature extraction and dual attention mechanisms, achieving superior long-term urban traffic prediction on real-world datasets. Wang et al. (2025) recently proposed a large language model (LLM)-driven probabilistic trajectory prediction framework, which integrates spatio-temporal encoding with normalizing flows to model uncertainty and complex motion dynamics. Although these methods struggle to represent the complex spatial posture changes during the module installation process, they show that accurate pose prediction of modules depends crucially on effective modeling of spatial relationships and temporal dependencies among trajectory points, particularly in complex movements. Spatial structures elucidate the distribution and interrelation of points in space, vital for deciphering movement patterns. Consequently, this paper introduces a DT-enabled module stacking system (DTPTS) with a novel network, PosTraFormer, which enhances controllability by modeling the complex spatial posture changes during the module installation process through multi-layer graph convolutional networks (GCNs), and by estimating posture and predicting trajectories through the integration of spatiotemporal modeling.

The lifecycle of MiC is illustrated in Fig. 1. Initially, dynamic spatiotemporal data from assets involved in the stacking process (such as cranes, workers, and modules) are captured using Ultra Wide Bands (UWBs) and Inertial Measurement Units (IMUs) for enhanced positioning accuracy in MiC environments. All real-time data and intrinsic characteristics, including geometry and semantics, are updated to the Smart Industry Foundation Classes (SIFCs) via the Mobile Gateway Operating System (MGOS) for efficient data management. DTPTS enables real-time automatic mapping of stacking activities and unified information sharing among stakeholders, while PosTraFormer analyzes key stacking factors for enhanced trajectory prediction. Our primary contributions are as follows:

(1) A DT platform based on IoT is proposed for real-time perception, pose estimation, and bidirectional interoperability between the cyber and physical modules in MiC stacking scenarios.
(2) A novel network called PosTraFormer is proposed to model the spatial–temporal structure of trajectory points to predict future motion trajectories, which is embedded into the DT platform to prevent potential structural collision and misaligned stacking.
(3) Comparative studies with the current state-of-the-art trajectory prediction models are conducted to validate the feasibility of the system and the effectiveness of the proposed method.

The remainder of this paper is structured as follows. Section 2 presents the research related to MiC. Section 3 describes the DTPTS framework. Section 4 gives the pose estimation and trajectory prediction algorithm. Comparative experiments are conducted in Section 5. Section 6 applies the DTPTS in a real-life MiC construction site and validates the effectiveness of our proposed method. Section 7 concludes the whole paper and gives the future research directions.

2. Literature review

This section reviews research related to MiC on DT, pose estimation, and trajectory prediction. At the end of this section, research gaps are summarized.

2.1. DT in MiC

DT has become a revolutionary force for MiC lifecycle management to facilitate real-time bidirectional interoperability between physical and virtual environments (Nguyen et al., 2025; Zhou, 2025). DT framework with four layers has been widely recognized by researchers as the conceptualized workflow which includes the physical layer (data acquisition), data layer (data transmission, integration, and preprocessing), virtual layer (digital modeling and data analysis), and service layer (visualization) (Tao and Zhang, 2017; Tuhaise et al., 2023; Pregnolato et al., 2022). In recent years, studies based on DT have made significant progress with the development of information and communication technology, such as high precision IoT devices, state-of-the-art artificial intelligence, robust cloud computing, secure blockchain, and scalable big data analytics (Rangasamy and Yang, 2024; Q. Liu et al., 2024). Representative papers can be roughly categorized into four aspects according to different MiC processes: (1) Design and planning; (2) Transportation scheduling; (3) Construction and management; (4) Operation and Maintenance.

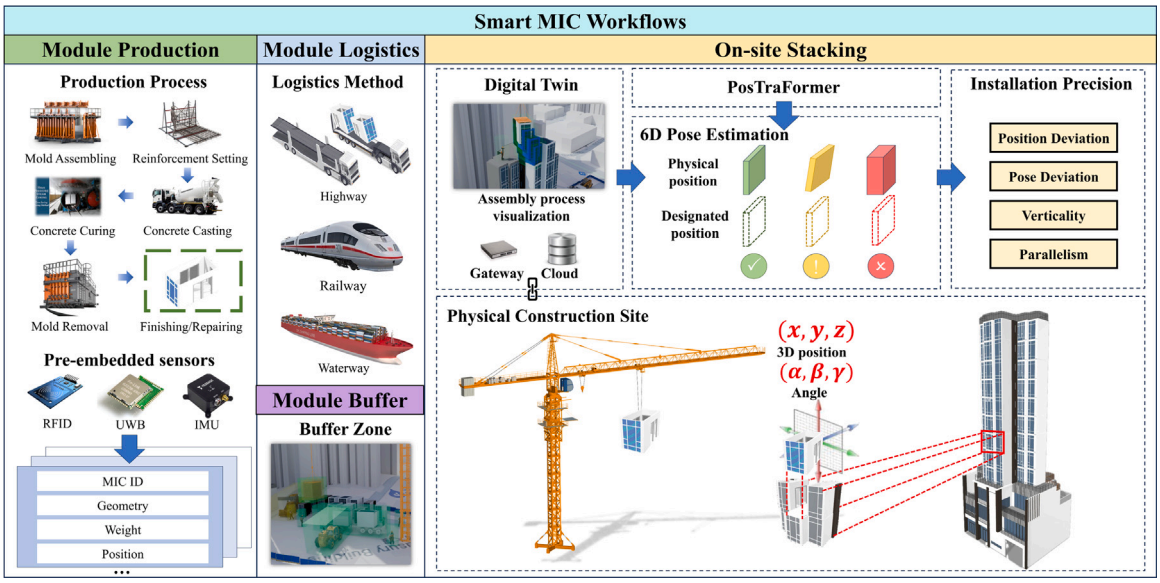


Fig. 1. Smart modular integrated construction lifecycle.

(1) **Design and planning.** During the design phase, DT technology can optimize MiC design schemes through three-dimensional modeling and simulation analysis under the unified digital design platform for multiple design stakeholders. Building Information Modeling (BIM) related 3D software has been extensively utilized to support the design generation and optimization for the MiC industry due to its advantages in the virtual representation of building geometries and semantics. For example, Li et al. (2024) integrated BIM and graphical convolutional neural networks to automatically generate prefabricated space layouts by extracting paired spatial geometric features. J. Liu et al. (2024) proposed a modular high-rise residential building design method based on graph constrained generative adversarial networks considering domain specific knowledge and constraints. Kosse et al. (2025) developed a modular semantic DT architecture to support knowledge integration and planning in construction projects. Li et al. (2025) proposed LSDTs, which integrate LLMs with DT to extract knowledge from unstructured documents and enable regulation-aware, high-fidelity, and adaptable infrastructure planning.

(2) **Transportation scheduling.** DT technology can eliminate transportation scheduling deviations by simulating logistics processes to keep synchronized with construction progress. Radio Frequency Identification (RFID) and Global Positioning System (GPS) are primarily used in most research for locating and recording the information of transport vehicles and modules and coordinating the supply chain in conjunction with on-site construction progress. Lee and Lee (Lee and Lee, 2021) developed a hybrid geographic information system framework with DT and IoT for real-time module logistics simulation and supply chain coordination, which predicts potential logistics risks and accurate module arrival times to reduce the module accumulation at the construction site. Tan et al. (2025) applied DT to prefabricated component factories, optimizing manufacturing and logistics processes.

(3) **Construction and management.** Hoisting safety is a critical research domain in prefabricated construction. Liu et al. (2021) proposed a safety risk coupling model using multiple data collection terminals and the apriori algorithm to mine associated risks, while Zhao et al. (2022) applied the Dijkstra algorithm for hoisting path planning with IMUs, GPS, and RFIDs, supported by LoRa-based data transmission. Beyond hoisting-specific studies, Jiang et al. (2020) developed an IFC-based BIM model for comprehensive safety management in mega projects, and Yuan et al. (2016) introduced a cyber–physical system to monitor temporary structures against failures. Other efforts have explored advanced human–machine interaction, such as Hasan et al. (2022), who linked augmented reality with DT to enhance equipment interaction. Building on this, DT technology has also been applied to MiC on-site assembly, where Jiang et al. (2022a,b) developed systems for real-time monitoring, planning, and execution using UWBs and RFIDs. More recently, Akhavian et al. (2025) proposed BIM2RDT, an AI-driven framework that converts BIM into a robot-ready DT to support automated operations and safety management.

(4) **Operation and maintenance.** In the maintenance process, DT technology enables intelligent management of the entire building lifecycle through real-time monitoring and predictive maintenance. A large number of studies about indoor comfort and energy conservation have been extensively reported. For instance, Tan et al. (2022) proposed a visualized DT lighting O&M system to support timely adaptive adjustment of ambient brightness considering pedestrians detected by YOLOv4. Hosamo et al. (2022) conducted a DT predictive maintenance framework for stable air handling unit functioning in heating, ventilation, and air conditioning (HVAC) systems in which analysis of variance and support vector machine are integrated for condition prediction. Tagliabue et al. (2021) focused on the indoor air quality problem and utilized IoT devices for real-time indoor air condition monitoring to control the HVAC systems and define opening/closing patterns for windows. Furthermore, Lu et al. (2020) proposed an anomaly detection system based on DT for asset monitoring which is based on IFC data integration. DT also supports emergency management by providing real-time monitoring and simulation drills for emergency plans, enhancing building safety and emergency response capabilities. For instance, Cheng et al. (2017) constructed a BIM based intelligent fire prevention system in which Bluetooth-based sensors and mobile apps are used for personal localization and relief guidance. Cheng et al. (2025) constructed high-fidelity multi-physics building models for multi-energy system analysis, enabling precise simulation and optimization of building operations. Additionally, DT technology has broad and effective applications in advanced project management (Pan and Zhang, 2021), structural health monitoring (Tuhaise et al., 2023), and information sharing (Lee et al., 2021).

2.2. Pose estimation

Pose estimation aims to accurately describe the position and rotation of an object in three-dimensional space. It has been widely used in many scenarios such as autonomous driving (Raffin et al., 2021), augmented reality (Song et al., 2018), and industrial robotics (Peng et al., 2022). To effectively improve efficiency in a complex, dynamic site environment with diverse equipment, many researchers have paid a lot of attention to the application in construction scenarios. These studies can be broadly categorized into two main types: non-visual sensor-based techniques (Azar et al., 2015) and vision-based techniques (Han and Lee, 2013; Han et al., 2013; Żywanowski et al., 2025).

Vision-based techniques use cameras to capture visual information, which can be further divided into marker-based and marker-less methods. For the marker-based method, tags are susceptible to damage due to harsh construction environments or equipment intense movement (Lundeen et al., 2016). In contrast, the marker-less methods are more suitable for the on-site environment of MiC. Liang et al. (2019) proposed a stacked hourglass network for marker-less excavator pose estimation to perform proximity detection and object tracking on construction sites. Tian et al. (2023) combined internet data and cinema4D generated data to construct a framework for real-time 3D position estimation of excavators. Recently, Fu et al. (2025) proposed a synthetic vision data-enhanced method for automated 3D pose recognition of construction workers, improving pose estimation accuracy and robustness in complex construction environments. Additionally, pose estimation can be found in other aspects of construction sites, including excavated objects, trucks, workers (Yu et al., 2021), and other equipment. However, these vision-based methods still cannot adapt to the complex dynamic environment of construction sites, as the accuracy of pose estimation is easily affected by variable and insufficient lighting conditions, object occlusion, and adverse weather conditions (Tang et al., 2023). Secondly, the coverage of construction scenes can only be achieved through the strategic placement of multiple cameras (Partsinevelos et al., 2024). Thirdly, rigorous system calibration at regular intervals is necessary to ensure precise camera alignment and the consistency of depth information. In such cases, the system needs to process a large amount of image data, consuming significant computational resources.

To address the shortcomings of vision-based pose estimation in construction, many researchers have increasingly focused on non-visual sensor-based technologies, which can be further categorized into single-sensor methods and sensor fusion methods. For example, Tang et al. (2022) proposed an IMU-based approach for unconstrained whole-body machines pose estimation in construction sites and provided the optimal IMU installation strategy. Liu et al. (2025) proposed the UMotion framework, which fuses six IMUs and UWB sensors using an Unscented Kalman Filter to achieve robust and accurate 3D human pose estimation. However, IMU cumulative errors may lead to the problem of pose drift over time. Methods based on GPS are prone to problems such as signal blocking and occlusion in the on-site assembly of MiC due to the lack of penetration into concrete (Groves, 2011). Meanwhile, the multipath effect can lead to large positioning errors. Radio frequency methods, such as Wireless Local Area Network (WLAN), UWBs, and RFIDs, lack orientation information for a single application. Compared with the limitations of single-sensor methods, the sensor fusion methods may provide a new solution for pose estimation in MiC on-site assembly, which allows for the integration of multiple sensors and the fusion of multimodal data. Although sensor fusion methods have achieved excellent results, the effects of lost data and signal interference should be considered, and further search for robust fusion algorithms to address calibration and synchronization between sensors should be made.

In conclusion, preliminary exploration has been conducted on the pose estimation for physical assets in construction sites. However, there is little research on module pose estimation for the on-site installation process. Through the literature review mentioned above, an innovative method based on sensor fusion can be found for module pose estimation to facilitate the correct alignment, connection, and positioning of modules. In addition, deep learning methods can help solve computational difficulties when fusing multimodal data due to their ability to efficiently process various types of data and extract valuable features (Brachmann et al., 2016; Cheng et al., 2021).

2.3. Trajectory prediction

The trajectory prediction aims to take a sequence of historical trajectories as input and outputs a predicted trajectory for a future period. Many researchers have integrated trajectory prediction into applications in architectural scenes, thereby reducing the occurrence of risk incidents. Trajectory prediction methods can be categorized into four types: physics-based methods, classical machine learning methods, deep learning methods, and reinforcement learning methods. Recently, data-driven methods are increasingly being utilized as they do not require explicit modeling of motion dynamics and can be generalized to various scenarios (Hu et al., 2020). Indeed, Long Short-Term Memory (LSTM) is one of the most widely used deep learning methods in trajectory prediction (Xue et al., 2018). Yang et al. (2023) proposed a framework for predicting worker trajectories based on LSTM networks, which comprehensively considers the interactions between workers and the interactions between workers and the environment. Hu et al. (2020) proposed an A* algorithm and dynamic window approach for finding the optimal path for construction robots and integrated the worker trajectory prediction method based on LSTM into the existing robot platform to enhance its performance potential. Zhang and Ge (2022) applied a trajectory prediction model based on the Transformer to predict the trajectories of loads and workers for the next ten seconds and established an early warning model. Further advancing deep learning for construction trajectory prediction, a novel framework employing a Transformer encoder-decoder with graph attention networks was developed to model worker temporal interactions and spatial relationships, achieving low error in short-term predictions (Alduais et al., 2025). However, they only considered the trajectories on the plane position and ignored the impact of the height during the object movement process. Nevertheless, several research gaps are identified from the literature review:

(1) DT technology for the module stacking process to enhance precision, coordination, and visualization is scarcely reported.
(2) Limited study explores multimodal data-based module pose estimation algorithms during MiC on-site stacking.
(3) Trajectory prediction in complex environments needs to be systematically analyzed.

Taking into account a certain spatial and temporal relationship of the trajectories of the lifting weights, this paper presents a DT-enabled pose estimation and trajectory prediction network based on spatio-temporal feature fusion modeling to fill these research gaps.

3. DT-enabled module stacking management system (DTPTS)

To overcome the challenges in MiC on-site stacking, this section proposes a DT framework to realize the interoperability of MiC stacking guidance, as shown in Fig. 2. The framework is divided into three layers: the physical layer, the virtual layer, and the service layer. The assets on site are embedded with smart IoT sensors for intelligent perception. Multidimensional and multiscale data of the static intrinsic properties and dynamic activities are captured by the IoT sensors. Then these data are uploaded to the IFC-based virtual layer via MGOS for the high-fidelity DT modeling. The service layer provides comprehensive stacking task management for project managers, stacking guidance for crane drivers, and collision warning for workers.

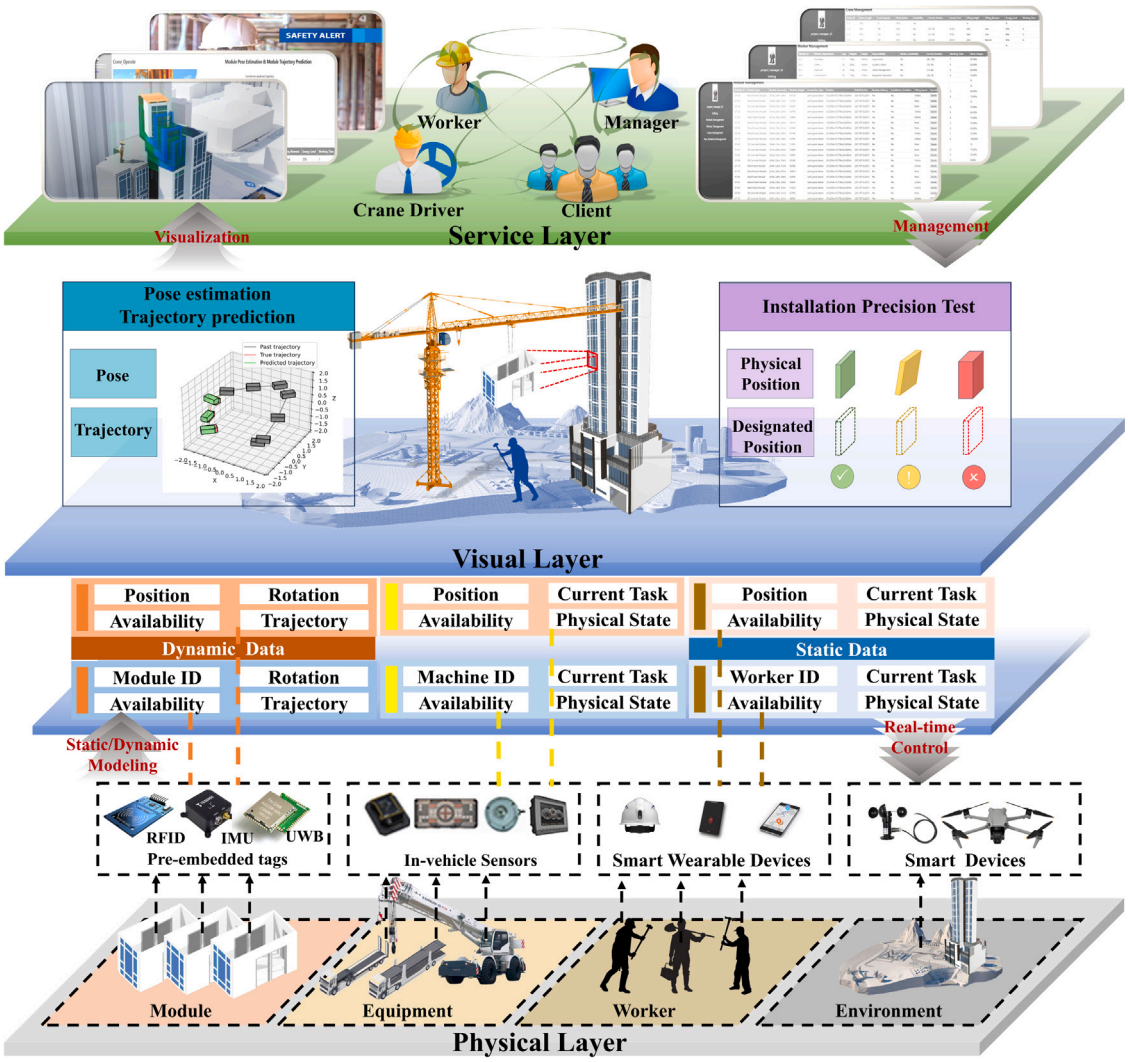


Fig. 2. The framework of DTPTS system.

3.1. Physical layer

The MiC stacking process is primarily composed of four main physical entities: assembly site environment, modules, equipment, and workers. On-site entities are embedded with intelligent IoT sensors to facilitate real-time data acquisition, dynamic perception, information sharing, and command acceptance for improving installation precision.

(1) **Assembly site environment.** Given the complexity and variability of actual assembly site, achieving a fully one-to-one replication within a digital environment for module stacking remains an unattainable objective. Therefore, critical environmental components that significantly impact module assembly are considered to ensure the DT model effectively balances the modeling complexity and actual assembly scenario mapping. The integration of two critical components, including static terrain and dynamic environmental changes, is essential. The terrain contours and existing road networks determine the layout of the buffer, crane, and modular building, as well as the transportation routes for module trailers. Surrounding permanent buildings influence the operational range of cranes. These static terrains are non-invasively captured by unmanned aerial vehicles equipped with cameras and laser scanners, supplemented by terrain data acquired from Google Earth Engine. Dynamic environmental changes, such as temperature, wind speed, and ground vibration, are captured in real-time by IoT sensors deployed on-site to mitigate the precision effect of pose estimation and trajectory prediction.

(2) **Modules.** Intelligent sensors integrating RFIDs, UWBs, and IMUs are embedded in modules before assembly to capture both static properties and dynamic poses. Globally unique identifiers, geometry, material types, weight, and connection methods are recorded in RFIDs. This information is crucial for module alignment, balancing the stacking center of gravity, selecting appropriate lifting equipment, and adjusting control parameters. To model spatio-temporal characteristics for pose estimation and trajectory prediction, accurate and real-time data acquisition of the module position and attitude is essential. UWB measures position with high precision and resistance to interference which is suitable for complex environments. However, UWB has limited capability in sensing rotational dimensions. IMU employs accelerometers and gyroscopes to provide high frequency and real-time measurements of acceleration and angular velocity, enabling precise tracking of rapidly changing dynamic postures. Nonetheless, IMU is prone to drift, which can lead to cumulative measurement errors and reduced localization accuracy. To address the inherent deficiencies of UWBs and IMUs, this study employs a Kalman filter to complementarily correct IMU drift and UWB's limited directional sensing capability. Additionally,

the selection of an appropriate number of base stations and optimization of their deployment positions are crucial to mitigate signal blocking, with the use of time difference of arrival method to enhance signal robustness.

(3) **Equipment.** The tower crane is the most crucial piece of machinery in the on-site assembly environment. Its static properties, such as jib length, maximum lifting capacity, maximum working radius, maximum load capacity, and maximum hook height, can be accurately obtained from technical handbooks and equipment nameplates. Dynamic activities of the tower crane are captured by crane-mounted sensors, such as load sensor, moment sensor, angle sensor, wind speed sensor, and acceleration sensor. Additionally, the tower crane integrates a human-machine interface that displays the module status in real-time and synchronized cyber–physical visibility. This interface also visualizes the pose estimation and trajectory prediction of the modules to guide crane operators in the stacking operations. Other construction equipment, such as module unloading forklifts, trailers, excavators, and trucks, are equipped with GPS for positioning to prevent potential collisions with on-site workers.

(4) **Workers.** Industrial wearables, including smart wristbands, smart helmets, and visualization interfaces, are equipped for workers. Static properties such as the worker's ID, department, age, weight, skill level, professional qualifications, and safety training records are stored in the smart wristband. Multi-functional IoT sensors in industrial wearables continuously update and monitor real-time location, working duration, heart rate, fatigue level, and relative distance to other entities.

(5) **MGOS.** LoRaWAN network communication protocol is used to define how IoT devices are connected and how data is uploaded. Star network architecture is adopted for simplified network management, maintenance, and expansion. MGOS updates the static properties and dynamic properties of the other four entities to the virtual layer in real time and transports control commands to the physical level for real-time interoperability.

3.2. Virtual layer

The virtual layer mainly realizes the unified management of the data uploaded by the physical layer and establishes the high fidelity DT of the stacking process.

3.2.1. Unified data management based on IFC

The multi-dimensional static and dynamic data in the stacking process are often uploaded and stored in different formats with different sensors. Manually extracting and processing diversified and fragmented stacking information is not only extremely time consuming but also hinders information sharing. Therefore, a unified standard building data architecture is needed to integrate and process fragmented data. IFC, as a widely adopted industrial data format, is considered for the unified DT construction and intelligent data management, as shown in Fig. 3. Firstly, the data of various aspects of the stacking process is organized with the "IfcRoot" entity as the root node to better describe the relationships and hierarchical structure. For objects involved in the MiC hoisting process, such as environment, workers, modules, and equipment, "IfcObjectDefinition" is used for definition, and the inherited description property is further explained for objects. This mainly includes static predefined properties and dynamic expandable properties. These static predefined properties come from the initial design model of the project and are visually constructed in 3D through Computer Aided Design (CAD) software (such as AutoCAD, Revit, and Solidworks). Dynamic properties refer to all dynamic activity information from the buffer zone to installation completion, which is collected in real-time by on-site IoT devices and pointed to the corresponding data position for updating and storage. The relationships between various objects are defined in detail using "IfcRelationship" to describe the bidirectional association. For instance, workers are usually responsible for installing, moving, or handling modules, and the "IfcRelAssignsToActor" can be used to associate modules with workers to describe which workers are responsible for specific modules. Cranes are typically used to move modules from the buffer zone to the installation position. The "IfcRelAssignsToCrane" relationship can be used to aggregate modules with cranes. Furthermore, IFC-compatible interfaces and application programming interfaces are developed to facilitate access and real-time modification of data by various types of software and visualization pages in the service layer. Real-time synchronization and consistency of data are ensured to support the interoperability of the DT.

3.2.2. Multi-dimensional high fidelity DT

The MiC stacking instance is a virtual mapping of physical stacking, synchronously inheriting the original parameters of physical stacking. Any changes in physical stacking will have real-time corresponding impacts on the respective MiC instances. DT modeling includes static and dynamic modeling. Dynamic modeling is the most complex stage, which will be comprehensive across the entire on-site assembly process and will need to consider a wide range of contingencies and inter asset interactions. It consists of four main components: (1) Dynamic environment modeling: Environmental factors are important for stacking accuracy, and sensors are used to accurately capture environmental variations (e.g., temperature, humidity, and windiness) and map them to the virtual environment. (2) Asset dynamic sensing: dynamic information of modules, workers, and cranes is transmitted to the cloud data via MGOS and reflected on the corresponding asset models in the virtual environment in real-time. (3) Asset interaction: The interaction and collaborative behaviors among assets are crucial for achieving installation accuracy. Three main interactions are considered in this section. The first one is the interaction between cranes and modules. The sensors of the crane and module need to provide their respective dynamic information at the same time stamp to facilitate the stacking adjustment based on the position of the module. The second one is the interaction between workers and cranes. The workers need to communicate with crane drivers in real-time to identify and deal with dynamic problems. The third one is the interaction between workers and modules. Workers need to assist in installing the module to the designated position and make necessary fixes and connections according to the module installation progress. (4) Emergency response: The DT system simulates and evaluates various possible risk scenarios, such as sudden change of wind, equipment failure, etc., as well as provides an emergency response plan. This helps to identify potential risks in advance and take responsive measures to mitigate losses.

3.3. Service layer

The service layer creates a real-time visualization and information sharing platform for all stakeholders together. It primarily encompasses two types of services:

(1) **Pose estimation and trajectory prediction for guidance service.** The on-site installation guidance service for construction workers and tower crane operators includes collision safety warnings and installation guidance respectively. The operators can adjust the crane tower in advance to avoid potential collisions based on the predicted trajectory of the module, which is presented in the DT enabled visualization platform. In addition, the crane operator can view the module stacking sequence list, as well as the real-time status (e.g. whether it has reached the buffer zone, whether it can be used for stacking) and real-time location of the module through the web page, which can significantly reduce the driver's time for searching the location of the module and the commanding officer's time for scheduling the stacking sequence. Construction workers on the ground

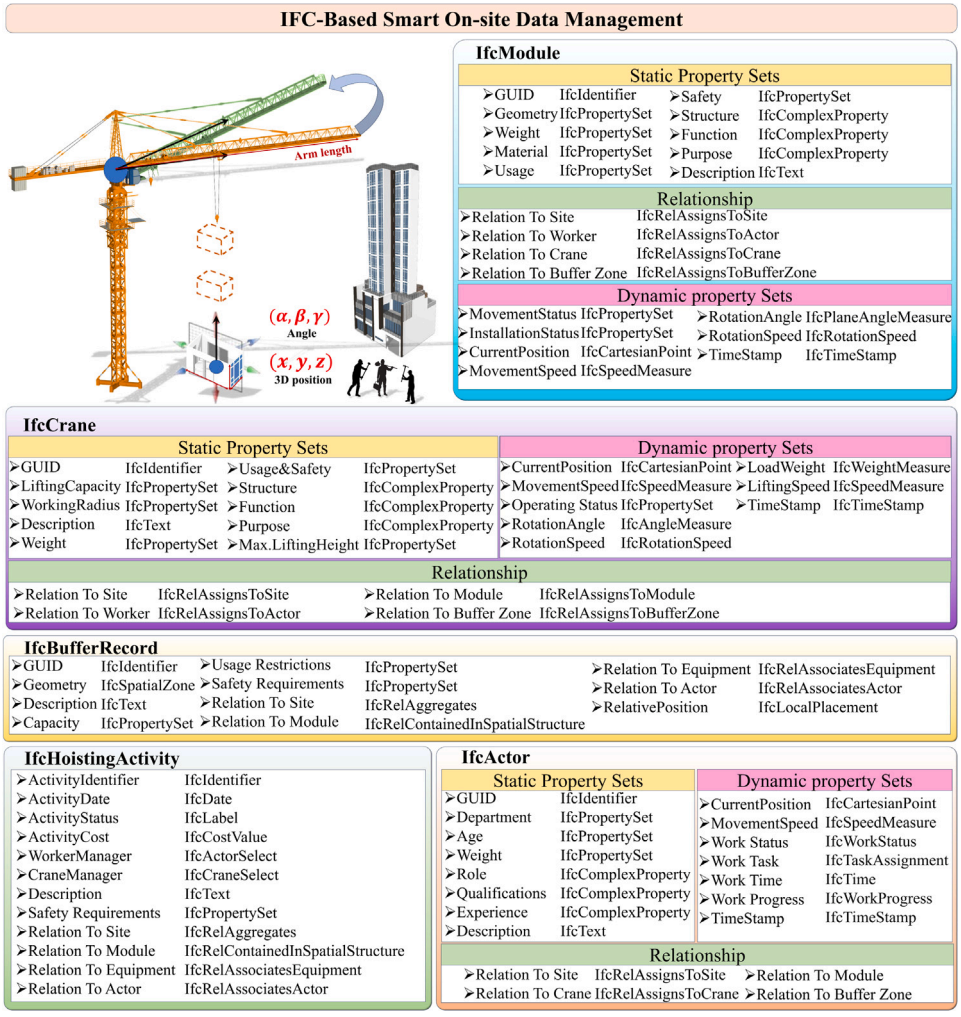


Fig. 3. IFC-based smart on-site data management.

assisting the crane driver can be informed of the safety of the area they are in by means of alarms installed on the wearable device, effectively avoiding potential falling objects and collision risks.

(2) **Management service.** The process management service for managers and clients covers process management, quality assurance, and safety management services. The relevant stakeholders have the right to access the dynamic map and verify the location information of prefabricated modules and workers at any time. Site managers and clients have access to the module installation progress web page, which provides a real-time indication of the status of the building's installation, represented by different modeling on the web page. The manager can track the construction progress in real time, which facilitates the scheduling of the subsequent installation process and the previous production and transportation stages.

4. PosTraFormer

The algorithm implementation mainly consists of data preparation, model training, and online testing phases, as shown in Fig. 4.

4.1. Pose estimation

Real-time pose estimation is a fundamental component of the following services. Module pose estimation is the estimation of the position and orientation of a module's coordinate system in the Earth's coordinate system. The complexity lies in the nonlinear nature of the orientation and the fact that the orientation can be parameterized in different ways (Kok et al., 2017). In this paper, the real-time 3D position information of intelligent modules is obtained by UWBs, and attitude solving is performed based on IMUs.

4.1.1. UWB distance measurement and positioning

The distance measurement method adopts one-sided bidirectional ranging. Node A sends a poll message and records the sending moment T^A_{poll}. Node B receives a poll message and records the receiving moment R^B_{poll}, then sends a response message and records the sending moment T^B_{resp}. Node A receives the response message and records the receiving moment R^A_{resp}. Therefore, the distance (transmission time) between the two points can be calculated using the above calculation process:

$$\hat{T}_{\text{prop}} = \frac{T_{\text{round}} - T_{\text{reply}}}{2} \tag{1}$$

$$T_{\text{round}} = R^A_{\text{resp}} - T^A_{\text{poll}} \tag{2}$$

$$T_{\text{reply}} = T^B_{\text{resp}} - R^B_{\text{poll}} \tag{3}$$

The coordinates of the module are calculated by the fixed base station. The coordinates of the fixed base stations are base station a (x_1, y_1, z_1), base station b (x_2, y_2, z_2), base station c (x_3, y_3, z_3), base station d (x_4, y_4, z_4). The coordinate of the desired positioning tag is

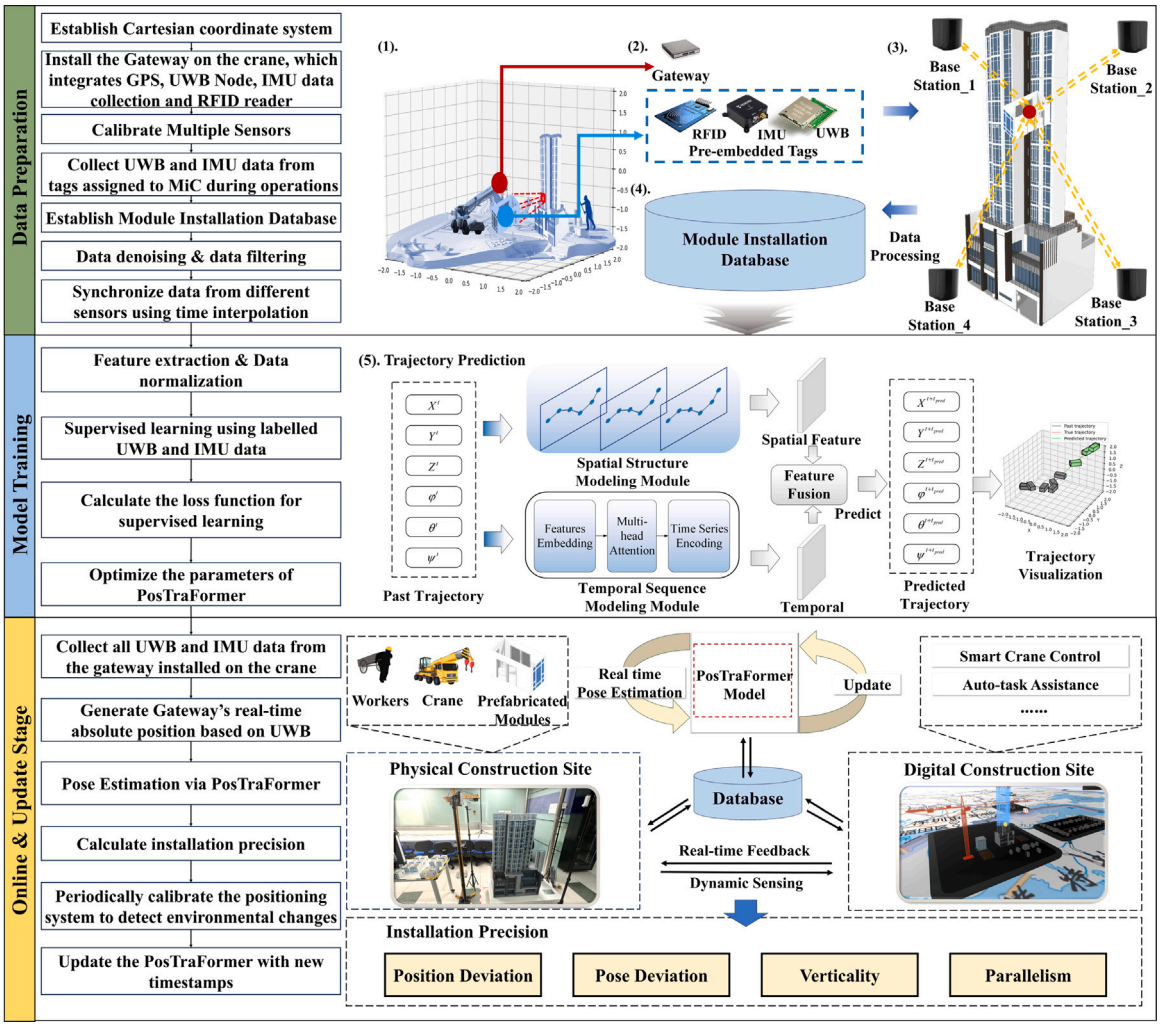


Fig. 4. Pose estimation and trajectory prediction algorithm.

O (x_0, y_0, z_0), which is calculated by the following formula:

$$\hat{T}_{\text{prop}}\ n \times v = \sqrt{(x_0 - x_n)^2 + (y_0 - y_n)^2 + (z_0 - z_n)^2} \tag{4}$$

where n is a positive integer taking values from 1 to 4.

4.1.2. IMU postural solution

In a high dynamic construction environment, IMU performs real-time attitude estimation using Euler angles to intuitively represent the object's orientation. These Euler angles include yaw (ψ), pitch (θ), and roll (ϕ), as illustrated in Fig. 5. Specifically, pitch (θ) is calculated based on the accelerometer readings. Using the normalized accelerations in the x and z directions, θ can be derived as:

$$\theta = \arctan\left(\frac{\text{acc}_x}{\sqrt{\text{acc}_y^2 + \text{acc}_z^2}}\right) \tag{5}$$

where acc_x, acc_y, and acc_z represent the accelerations along the x, y, and z axes, respectively.

Roll (ϕ) is computed using the accelerations along the y and z axes, as follows:

$$\phi = \arctan\left(\frac{-\text{acc}_y}{\text{acc}_z}\right) \tag{6}$$

The yaw angle (ψ) is determined using the magnetometer readings, after compensating for pitch and roll. The corrected yaw is given by:

$$\psi = \arctan\left(\frac{\text{mag}_y \cdot \cos\phi - \text{mag}_z \cdot \sin\phi}{\text{mag}_x \cdot \cos\theta + \text{mag}_y \cdot \sin\theta \cdot \sin\phi + \text{mag}_z \cdot \sin\theta \cdot \cos\phi}\right) \tag{7}$$

where mag_x, mag_y, and mag_z represent the magnetometer readings along the x, y, and z axes, respectively.

In construction module pose estimation, errors often arise due to sensor noise, drift, and environmental interference. IMUs can experience drift, especially in gyroscopes, while magnetometers are affected by local magnetic disturbances. Additionally, mechanical vibrations, sensor misalignment, and time synchronization issues between multiple sensors further contribute to inaccuracies in the module's orientation estimation. When deviations between the actual and ideal values occur, the accelerometer is used to correct deviations in the pitch and roll angles. First, the current accelerometer outputs are normalized:

$$\mathbf{acc} = \frac{\mathbf{acc}}{\|\mathbf{acc}\|} \tag{8}$$

where **acc** is the vector of accelerations in the x, y, and z directions.

The ideal accelerometer output, in a perfect navigation coordinate system, should only include the gravitational component. Therefore, the normalized ideal acceleration in the navigation coordinate system is:

$$\mathbf{acc}_{\text{ideal}} = [0, 0, 1] \tag{9}$$

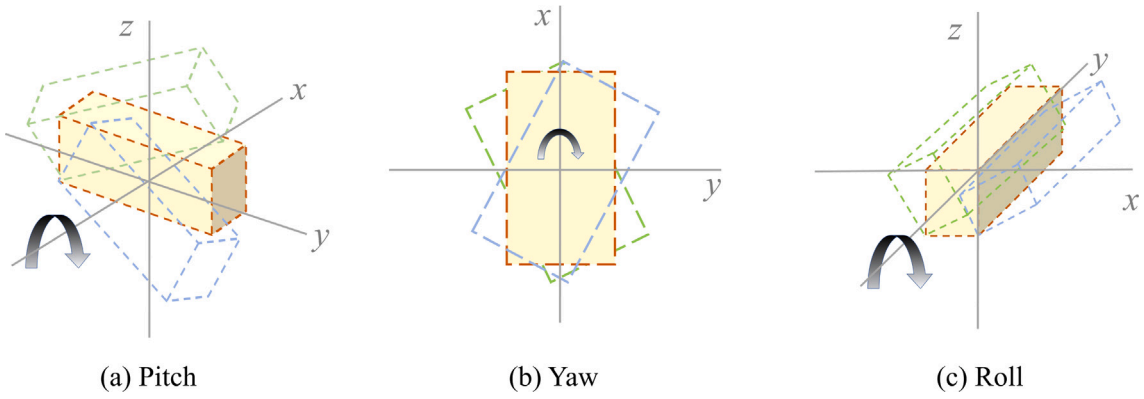


Fig. 5. Euler angles for modular poses.

The deviation between the actual and ideal values is calculated using the cross product:

$$\Delta_{\text{pitch,roll}} = \mathbf{acc} \times \mathbf{acc}_{\text{ideal}} \tag{10}$$

This deviation is then used to correct the pitch and roll angles by applying a small adjustment. Similarly, the magnetometer is used to correct deviations in the yaw angle. Like the accelerometer, the magnetometer outputs are normalized:

$$\mathbf{mag} = \frac{\mathbf{mag}}{\|\mathbf{mag}\|} \tag{11}$$

The magnetometer output is then transformed into the navigation coordinate system:

$$\mathbf{mag}_{\text{nav}} = \mathbf{C}_{\text{n}}^{\text{b}} \cdot \mathbf{mag} \tag{12}$$

where $\mathbf{C}_{\text{n}}^{\text{b}}$ is the direction cosine matrix (rotation matrix) from the body frame to the navigation frame.

The deviation between the actual and ideal magnetometer outputs is again calculated using the cross product:

$$\Delta_{\text{yaw}} = \mathbf{mag}_{\text{nav}} \times \mathbf{mag}_{\text{ideal}} \tag{13}$$

This deviation is used to adjust the yaw angle accordingly.

4.2. Trajectory prediction algorithm

4.2.1. Problem description

Let X represent all trajectories, including the installation trajectories of N prefabricated modules. It can be denoted as $X = \{X_1, X_2, \ldots, X_N\}$. The time step is $1, 2, \ldots, t_{obs}$, t_{obs} denotes the length of the observed sequence. Module 6D trajectory prediction uses historical data to predict trajectories over time. The past trajectory of module i is represented as $X_i = (x_i^t, y_i^t, z_i^t, \varphi_i^t, \theta_i^t, \psi_i^t)$, and $x_i^t, y_i^t, z_i^t, \varphi_i^t, \theta_i^t, \psi_i^t$ respectively represents x coordinate, y coordinate, z coordinate, roll, pitch and yaw of the module at time t. When the time step is $T = (t_{\text{obs}} + 1, t_{\text{obs}} + 2, \ldots, t_{\text{obs}} + t_{\text{pred}}, \ldots)$, the true trajectory of module i is:

$$Y_i = \left(x_i^{t+t_{\text{pred}}}, y_i^{t+t_{\text{pred}}}, z_i^{t+t_{\text{pred}}}, \varphi_i^{t+t_{\text{pred}}}, \theta_i^{t+t_{\text{pred}}}, \psi_i^{t+t_{\text{pred}}}\right). \tag{14}$$

The predicted trajectory is:

$$\hat{Y}_i = \left(\hat{x}_i^{t+t_{\text{pred}}}, \hat{y}_i^{t+t_{\text{pred}}}, \hat{z}_i^{t+t_{\text{pred}}}, \hat{\varphi}_i^{t+t_{\text{pred}}}, \hat{\theta}_i^{t+t_{\text{pred}}}, \hat{\psi}_i^{t+t_{\text{pred}}}\right) \tag{15}$$

where $\hat{x}_i^{t+t_{\text{pred}}}, \hat{y}_i^{t+t_{\text{pred}}}, \hat{z}_i^{t+t_{\text{pred}}}, \hat{\varphi}_i^{t+t_{\text{pred}}}, \hat{\theta}_i^{t+t_{\text{pred}}}, \hat{\psi}_i^{t+t_{\text{pred}}}$ respectively represent the predicted x coordinate, y coordinate, z coordinate, roll, pitch and yaw of the module at future time $t_{\text{obs}} + t_{\text{pred}}$.

4.2.2. Spatial structural relationship modeling

In a set of trajectory points $X = \{X_1, X_2, \ldots, X_N\}$, there exists an inherent spatial structure relationship between the points, which is crucial for accurately predicting the subsequent trajectory points. Traditional methods may overlook these intricate spatial dependencies, leading to suboptimal predictions. To address this, we employ a graph model to capture the spatial relationships among the trajectory points, enabling a more comprehensive understanding of the underlying patterns. As shown in Fig. 6, the trajectory points are encoded into a graph structure, where each point X_i is represented as a node in the graph. The edges between these nodes, which signify the spatial relationships, are defined by an adjacency matrix A. Unlike fixed adjacency matrices, we utilize an adaptive approach where A is a learnable parameter matrix. This allows the model to dynamically adjust the connections between nodes, thus better capturing the complex spatial dependencies present in the data. The encoded trajectory points are then fed into an adaptive GCN with multiple layers. The graph convolution operation at each layer can be expressed as:

$$H^{(l+1)} = \sigma\left(\tilde{A}H^{(l)}W^{(l)}\right) \tag{16}$$

where $H^{(l)}$ is the feature matrix of the trajectory points at layer l, $\tilde{A}$ is the normalized adaptive adjacency matrix, $W^{(l)}$ is the trainable weight matrix, and σ is a non-linear activation function. The initial feature matrix $H^{(0)}$ corresponds to the input trajectory points. Through multiple layers of graph convolution, the model iteratively refines the feature representations, enabling the extraction of spatial structure features: $S_f = H^{(l+1)}$. S_f is then used to predict the next set of trajectory points. The adaptive nature of the adjacency matrix ensures that the model can effectively learn and capture the spatial dependencies, leading to more accurate trajectory predictions.

4.2.3. Time series relationship modeling

The trajectory points not only present a certain distribution pattern in space, but also show sequentiality and dependence in the time dimension. The forward backward temporal relationship reflects the continuity, speed change, and direction trend of the object movement, and ignoring temporal information may lead to misjudgment of the future trajectory. To effectively model these temporal relationships, we employ a multi-head attention mechanism, which allows the model to simultaneously focus on different aspects of the sequence, capturing both short term and long term dependencies. The time series relationship modeling module architecture is shown in Fig. 6. The input size is $B \times F \times L$, where F is the length of the input features, B is the batch dimension, and L is the sequence length. In this study, the input is the normalized three-dimensional coordinates and the deflection attitude, so F equals 6. The normalization method is 0 to 1 normalization. Firstly, two trainable linear layers are used to map the inputs into high dimensional features. Subsequently, PosTraFormer utilizes positional embedding to retain the positional information of the sequence. The

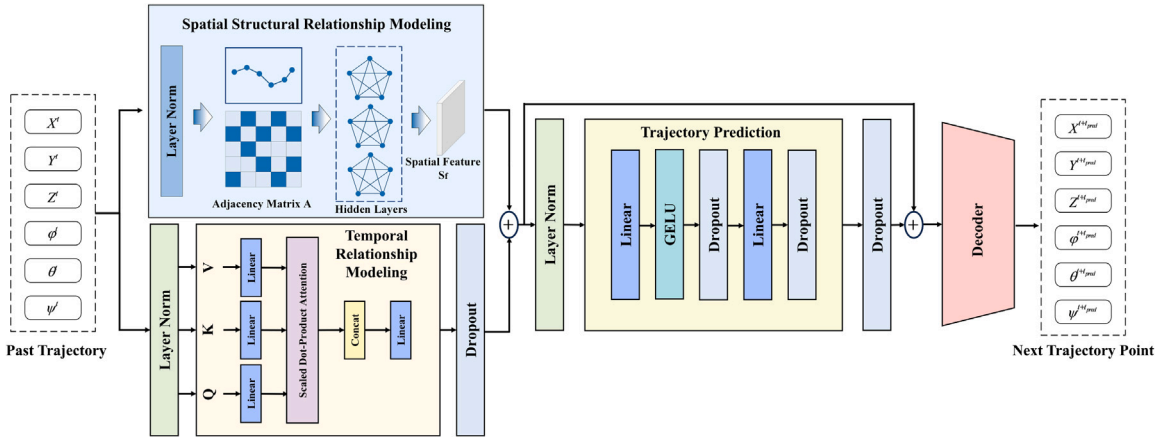


Fig. 6. The overall framework of PosTraFormer.

historical trajectory is encoded by positional information, combined with the relative positions between trajectory points. Position Encoding (PE) is as follows:

$$PE(pos, 2i) = \sin\left(\frac{pos}{10000^{\left(\frac{2i}{d_{\text{model}}}\right)}}\right)$$
$$PE(pos, 2i+1) = \cos\left(\frac{pos}{10000^{\left(\frac{2i}{d_{\text{model}}}\right)}}\right) \tag{17}$$

where pos represents position, i denotes the dimension, and d_{model} signifies the dimensionality of the model. In this paper, d_{model} is 512.

$$Z = \left[X_1 E, X_2 E, \ldots, X_N E\right] + PE \tag{18}$$

After embedding by a linear mapping $E \in \mathbb{R}^{F\times C}$ and summing with the positional embedding $PE \in \mathbb{R}^{F\times C}$, the input sequence is sent to the encoder of PosTraFormer. where C is the embedding dimension. The PosTraFormer encoder consists of a multi-head self-attention module and a feedforward module. Scaled dot-product attention is the core functionality of PosTraFormer. It is described as a mapping function that uses the query matrix Q, the key matrix K, and the value matrix V to compute the correlation between different positions in the input sequence and maps it to the output attention matrix. This allows the model to capture the importance and correlation between locations in the input sequence.

$$\text{Attention}\ (Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V \tag{19}$$

where the query matrix Q is dot product with the K matrix. The scale factor $\sqrt{d_k}$ is used for proper softmax normalization. Thus, preventing very small gradients when large values of d cause the dot product to grow significantly.

MSA parallelizes the semantic information of the input sequence from different positions. A richer and more expressive feature representation is obtained by splicing the weighted results of multiple attention heads.

$$MSA(Q, K, V) = \text{Concat}\left(H_1, \ldots, H_m\right) W \tag{20}$$

$$H_i = Attention\left(Q_i W_i^Q, K_i W_i^K, V_i W_i^V\right), i \in [1, \ldots, m], \tag{21}$$

where $W_i^Q \in \mathbb{R}^{d_{\text{model}}\times d_k}, W_i^K \in \mathbb{R}^{d_{\text{model}}\times d_k}, W_i^V \in \mathbb{R}^{d_{\text{model}}\times d_v}$, $W \in \mathbb{R}^{hd_k\times d_{\text{model}}}$.

4.2.4. Trajectory prediction via spatio-temporal feature fusion

To enhance the accuracy of trajectory prediction, we integrate the spatial features S_f extracted from the spatial structure modeling module with the temporal features H_f obtained from the multi-head attention mechanism. This fusion leverages both the spatial dependencies among the trajectory points and their sequential temporal relationships. The combined feature representation captures a comprehensive understanding of the trajectory dynamics. The fusion process involves linear concatenating S_f and H_f, which can be expressed as:

$$Z = \lambda \cdot S_F + (1-\lambda) H_F \tag{22}$$

where λ is a trainable weight coefficient to control the influence of spatial and temporal features after fusion.

The fused spatio-temporal features are next used to predict the location of future trajectory points. Position wise feed forward is the feed forward neural network processing of the hidden representation of each position. Specifically, the module applies two linear layers and the ReLU activation function to nonlinearly transform the position information. This helps the model to better capture the semantic information of each position when processing sequence data, and to provide a rich representation of the position information while keeping it constant. All the above components form the Encoder block, which is stacked appropriately to form the Encoder. This encoded information is input into the encoder for feature learning, generating intermediate encoded information, which is then transmitted to the decoder for decoding.

$$Z_\iota' = LN\left(MSA\left(Z_{\iota-1}\right) + Z_{\iota-1}\right), \iota = 1, 2, \ldots, L \tag{23}$$

$$Z_\iota = LN\left(PWFF\left(Z_\iota'\right) + Z_\iota'\right), \iota = 1, 2, \ldots, L \tag{24}$$

$$Y = \text{LN}\left(Z_\iota\right) \tag{25}$$

where $LN(\cdot)$ denotes the layer normalization operator, + denotes the residual join, and L denotes the number of encoder block stacks.

The Decoder block and the Encoder block have similar structures. But different from the Encoder block, two MSA layers are included in the Decoder block. During training, the model applies MSA layers to prevent attention to future tokens, ensuring that each token can only notice preceding tokens.

4.2.5. Loss function

Trajectory prediction is a regression task, and the model's loss is optimized using Smooth L1 Loss:

$$\text{SmoothL1} = \frac{1}{n}\sum_{i=1}^{n}\begin{cases} 0.5 * \left(p_{i_p}^t - p_{i_{GT}}^t\right)^2, \text{ if } \left|p_{i_p}^t - p_{i_{GT}}^t\right| < 1 \\ \left|p_{i_p}^t - p_{i_{GT}}^t\right| - 0.5, \text{otherwise} \end{cases} \tag{26}$$

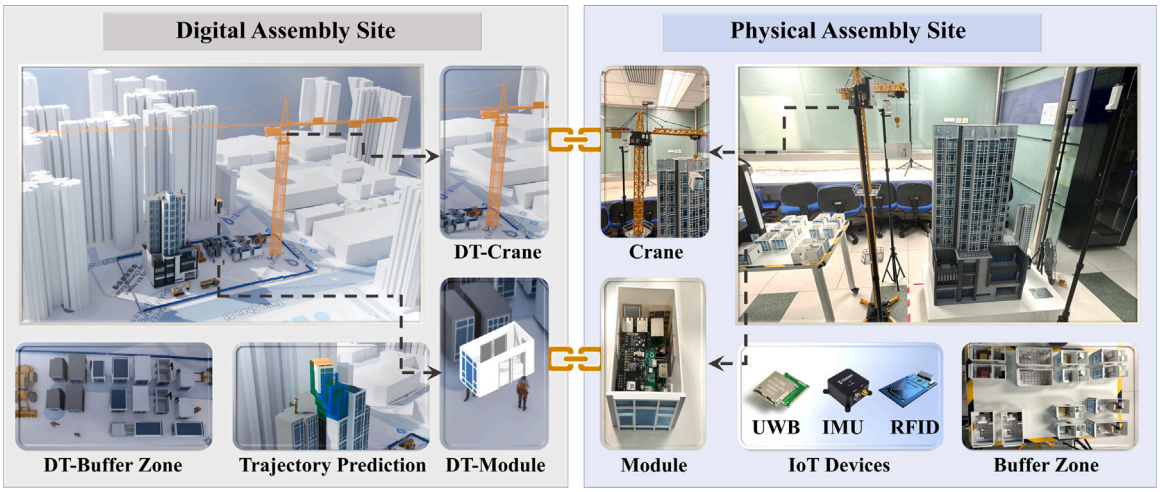


Fig. 7. The physical simulation experimental platform.

where $p^t_{i_p}$ and $p^t_{i_{GT}}$ represent the predicted and ground truth positions and poses. Within the interval of $\left|p^t_{i_p} - p^t_{i_{GT}}\right| < 1$, Smooth L1 Loss adopts the square function from L2 loss, addressing the issue of nondifferentiability at zero in L1 loss. Beyond the interval of $\left|p^t_{i_p} - p^t_{i_{GT}}\right| > 1$, it employs the linear function from L1 loss, mitigating the problem of gradient explosion for outliers. Consequently, Smooth L1 Loss maintains good stability and robustness when dealing with errors of varying magnitudes.

5. Experiments

5.1. Experimental setting

5.1.1. Experimental scenarios and data preparation

To verify the feasibility of the DTPTS an experimental platform for hoisting pose estimation and trajectory prediction is conducted in a real laboratory setting, as shown in Fig. 7. The setup comprises a physical stacking site, a digital stacking site, and an intelligent data management system. The physical stacking site mainly includes a buffer zone for storing modules, a building under construction, cranes, modules, workers, gateway, and base stations. The buffer zone measures 0.7 m in length and 0.35 m in width, capable of storing up to 16 modules. The building model follows the mainstream Podium System construction method, standing 1.35 m tall with 15 floors, including a podium section at the bottom and a framework section above. Base stations are strategically deployed around the building. Each room is modular and demountable, closely resembling real construction installation scenarios. Additionally, every module embeds intelligent RFID tags, UWBs, and IMUs. The smart crane has an operating radius of 1 m, a maximum payload of 10 kg, and a lifting speed of 0.1 m/s. A gateway is mounted on the crane hook to capture real-time data from modules and transmit control commands to the tower crane. The digital stacking site is developed by WebGL, which serves as the DT of the aforementioned physical assets. To facilitate unified management of diverse structural data, a "Trinity" platform is developed, encompassing intelligent modules, workers, and cranes.

The entire trajectory from hoisting the module in the buffer zone to placing it at the designated position is sampled at 2 Hz. A sliding window of fixed length is used to trim the trajectory starting from every alternate time step of the original trajectory (specifically, 5 s, 10 frames). Each trimmed trajectory is divided into two parts: the historical trajectory (input) and the predicted trajectory (output or label), totaling 1200 trimmed trajectories. In the experiment, the dataset is split into training and evaluation sets in a 7:3 ratio. The training set is utilized to optimize the parameters of the trajectory prediction model, while the evaluation set monitors the convergence of the model. Additionally, the module storage and movement status, construction progress of each building layer, and occupancy status of the tower crane are automatically updated at the same timestamp during this process. Dynamic activity data of workers, such as their movement paths and actions on the construction site, are captured via smart wristbands and helmets. All the aforementioned data is integrated into the SIFC database and subjected to data calibration and synchronization to ensure consistency and accuracy among various data sources. To address possible jitter and anomalies in the collected data, Kalman filtering is employed for smoothing and noise removal. Polynomial interpolation is utilized to handle missing values. Finally, to standardize the scale of position and orientation data for model training, the data is normalized.

5.1.2. Data processing

In this section, we describe the process of augmenting the lifting dataset through data augmentation techniques to ensure that various possible lifting scenarios are covered during model training, including normal lifting trajectories as well as anomalous situations (e.g., sensor malfunctions). As a result, the dataset size was expanded from a few hundred trajectories to 6000 trajectories. The data augmentation methods primarily include two techniques: using Time-Series Generative Adversarial Network (TimeGAN) to simulate real lifting data and injecting anomalous trajectory points into certain trajectory data using noise.

Real normal stacking data simulation. TimeGAN is used to generate samples that resemble the distribution of real lifting data. The generator of TimeGAN learns the temporal patterns of the real data, thus simulating new trajectory data. This approach effectively enhances the diversity of the dataset, especially when the original data is limited. Specifically, we input real lifting data into the TimeGAN model and train the generator to produce trajectory point sequences similar to the real data. After several iterations, the generator can output a large amount of new data that is temporally similar to the real data. To validate the effectiveness of the generated data, we compared the distribution of the generated data with the real data using principal component analysis. Fig. 8 demonstrates that the generated data's distribution in the two-dimensional space closely matches the distribution of the real data. These generated data not only increase the sample size but also enhance the diversity of the dataset, helping to improve the model's generalization ability, particularly when dealing with unknown lifting situations.

Anomalous stacking data simulation. In addition to using TimeGAN to generate normal lifting data, we also simulated anomalies by

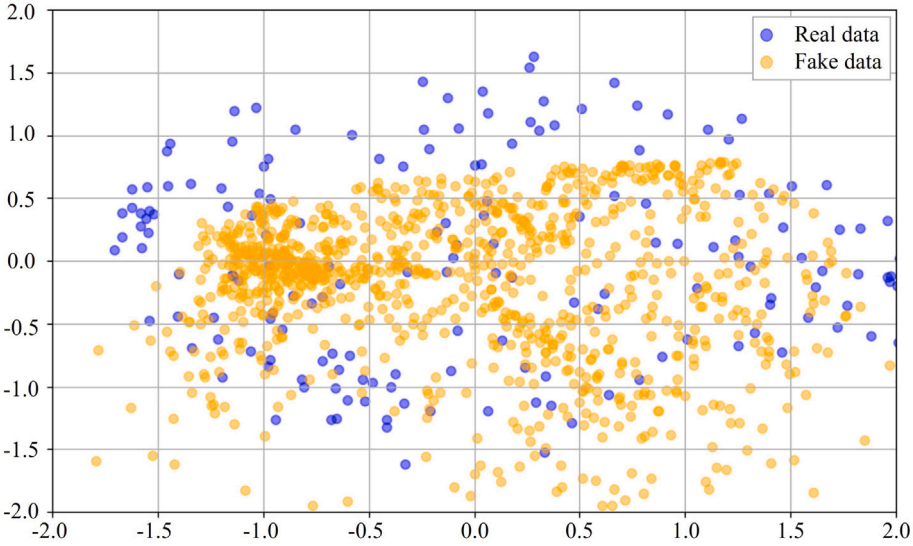


Fig. 8. Comparison of data distribution between simulated data and real data.

inserting noise points into some of the trajectory data. These anomalous trajectory points represent potential faults or irregular events that may occur during actual lifting operations, such as sensor malfunctions, data loss, or irregularities in lifting procedures. Specifically, we randomly inserted Gaussian noise at certain positions in the original trajectory data to create new trajectories with anomalous points. To ensure that the generated anomalous data still maintains some physical plausibility, the positions and magnitudes of the noise insertion were constrained, ensuring that the anomalous points neither completely distort the data nor introduce unrealistic situations, while still simulating some potential actual anomalies. For example, when simulating a sensor malfunction, the inserted noise might cause a sudden change in part of the data, while when simulating irregularities in the lifting operation, the noise points might cause discontinuities or fluctuations in the trajectory curve. These trajectories with anomalous points further increase the diversity of the dataset and help train the model to better handle unexpected situations that might arise during actual lifting operations.

5.1.3. Data diversity analysis

The lifting trajectory dataset was collected across a variety of lifting operation scenarios, including tasks such as module lifting, descending, translation, and rotation under different load conditions. Each type of lifting task exhibits distinct trajectory characteristics depending on the environmental conditions. To ensure sufficient diversity in the dataset, we not only increased the dataset size during the augmentation process but also specifically accounted for anomalous situations within the data, such as sensor malfunctions, data loss, and irregularities in lifting operations, thus ensuring that the dataset covers a wide range of potential lifting scenarios. To further analyze and visualize the diversity of the augmented dataset, we present the probability density distribution of the enhanced data in Fig. 9. The multiple peaks in this figure illustrate the diversity within the data feature distribution. Each peak corresponds to a region where data is concentrated in a particular scenario, and the presence of multiple peaks indicates that the data exhibits distinct features across different types of lifting operations. This diversity ensures that the model can learn the complex dynamics involved in lifting operations and is well equipped to adapt to potential anomalous situations that may arise during actual operations.

5.1.4. Evaluation metrics

The paper chooses Final Displacement Error (FDE) and Average Displacement Error (ADE) as model evaluation metrics, as they are the most widely used evaluation metrics in the field of trajectory prediction research.

(1) FDE: This is the Euclidean distance between the last point of the predicted trajectory and the last point of the real trajectory. It measures the long-term accuracy of the model's prediction.

$$FDE = \frac{1}{n}\sum_{i=1}^{n}\left|p_{i_p}^{tf} - p_{i_{GT}}^{tf}\right| \tag{27}$$

(2) ADE: This is the average Euclidean distance between all points of the predicted trajectory and the corresponding points of the real trajectory. It measures the overall accuracy of the model's prediction.

$$ADE = \frac{1}{n}\sum_{i=1}^{n}\frac{1}{t_f}\sum_{t=1}^{t_f}\left|p_{i_p}^{t} - p_{i_{GT}}^{t}\right| \tag{28}$$

In Eqs. (27) and (28), $p_{i_p}^{t}$ and $p_{i_{GT}}^{t}$ represent the predicted and ground truth positions and poses, respectively, at the final time step t_f for the $i-th$ trajectory and the t_f represents the prediction time length. It is worth noting that both FDE and ADE compute the total prediction error after normalization of angles and positions in the module stacking process, so these two evaluation metrics have no units.

5.2. Prediction frame selection

Before comparative experiments, the ratio of past trajectories and future trajectories is tested to determine the optimal ratio of input to output of the model. The different ratios are taken as 3:7, 4:6, 5:5, 6:4, 7:3, and 8:2 for experiments under the same dataset, respectively. The evaluation metrics are shown in Table 1 and the box plots for each evaluation metric are obtained as shown in Fig. 10.

From Fig. 10, when the ratio of input to output is 3:7, the median and average values of the evaluation metrics of the model are the highest, which means that the 3D positions and poses of the predicted trajectories have the highest gap with the real trajectories. In Table 1, it can also be seen that the model is most ineffective at 3:7, with FDE and ADE being the highest of all ratios at 0.1335 and 0.2011, with FDE reaching 6.39 times the optimal value, and ADE reaching 5.28 times the optimal value. When the ratio of input to output is 4:6 and 5:5, the model has more outliers and is less robust. As the ratio of past trajectories increases, the robustness of the prediction model increases while the prediction error gradually decreases. This observation is consistent with our intuitive understanding that greater visibility into past trajectories leads to more accurate predictions of future trajectories. The median and mean of the model's evaluation matrices are both minimized when the ratio of input to output is 8:2, which indicates that the 3D position and attitude of the predicted trajectories have the smallest gap with the real trajectories, and the

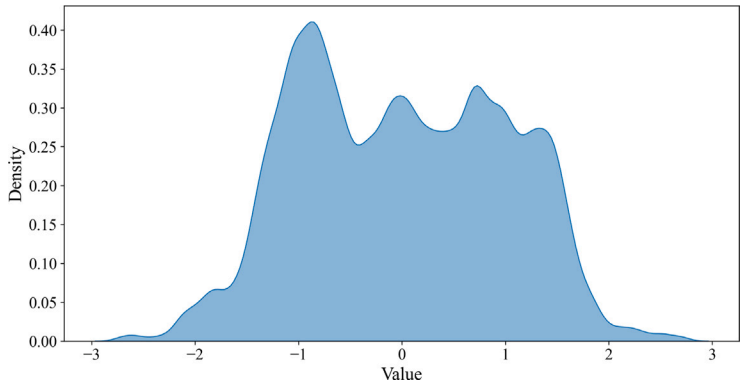


Fig. 9. Probability distribution of the data set.

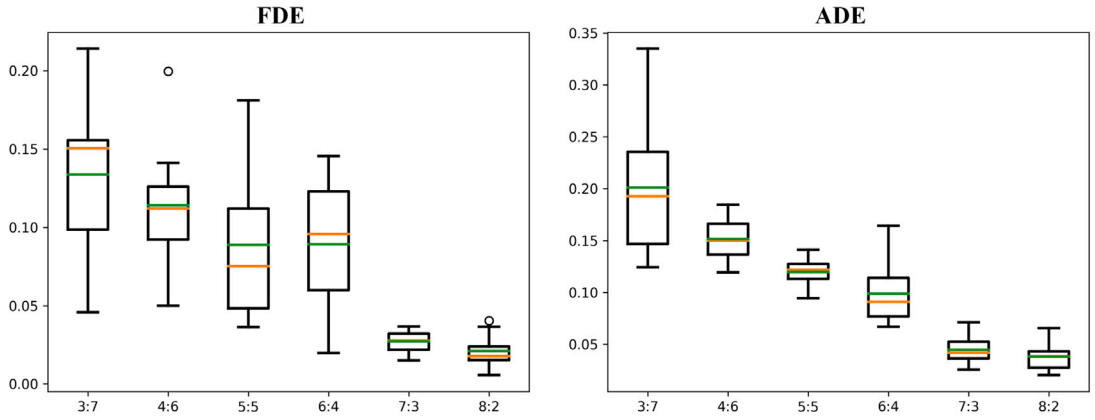


Fig. 10. The boxplot for different ratios of input to output. The green line indicates the average, which is affected by all data points, is very sensitive to outliers, and characterizes the overall average of the data. The orange line indicates the median, which is insensitive to outliers and can effectively reflect the central tendency of asymmetric data. Circles are data outliers and characterize anomalous data.

Table 1
The evaluation of different ratios of input to output.

Ratios of input to output	FDE	ADE
3:7	0.1335	0.2011
4:6	0.1140	0.1515
5:5	0.0887	0.1195
6:4	0.0891	0.0987
7:3	0.0271	0.0446
8:2	**0.0209**	**0.0381**

Table 2
Comparison of model performance on training and test datasets.

Dataset	Loss	FDE	ADE
Training	0.0269	0.0271	0.0446
Testing	0.0308	0.0313	0.0477

Table 3
Anomalous trajectory prediction results.

Sample	FDE	ADE
Normal	0.0304	0.0317
Abnormal	0.0321	0.0332

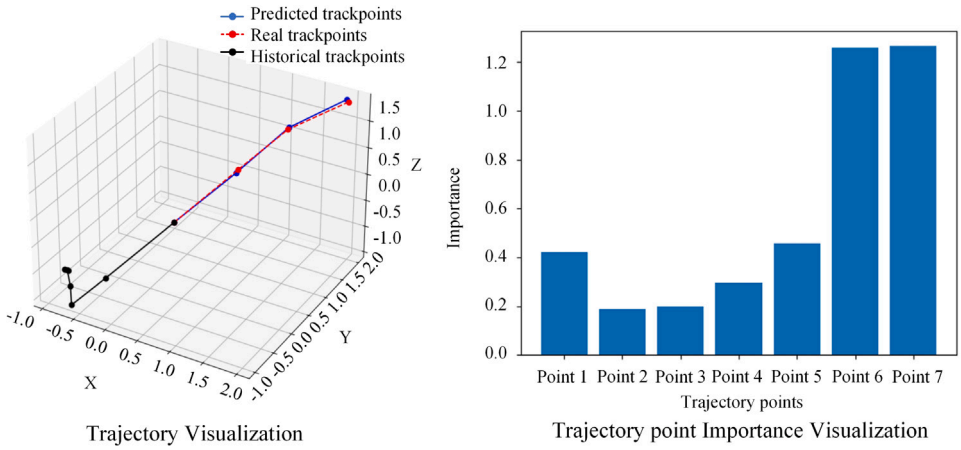
same can be seen in Table 1. However, compared with the model performance at the 7:3 ratio, 8:2 does not show significant advantages and seriously impairs the comprehensiveness of the access to future information, affecting the driver's ability to obtain real-time trajectory information and adjust the module accordingly. Therefore, considering the practical needs and model performance, we chose the 7:3 ratio for the subsequent study.

5.3. Model generalization analysis

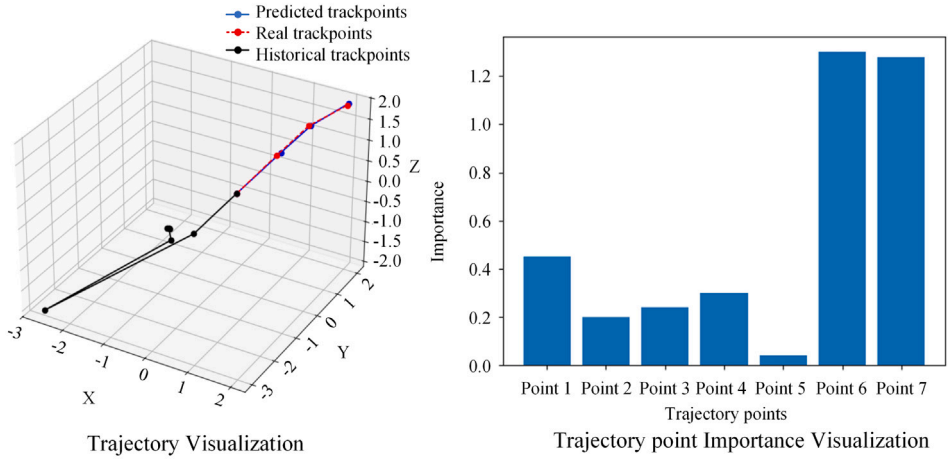
To validate the generalization ability of the model, we conducted a comparative analysis of its performance on both the training and test datasets, examining whether the model can effectively learn generalizable patterns from the training data and perform well on external data. Table 2 presents the average loss (FDE and ADE) on both the training and test datasets. The results show that the model not only achieves good performance on the training data but also performs equally well on the test data. The introduction of augmented data ensured that the features learned by the model are not merely local patterns from the training data, but rather are capable of adapting to diverse stacking scenarios.

5.4. Anomalous trajectory performance prediction analysis

Table 3 presents the model's prediction performance on samples containing anomalous trajectory points, including metrics (ADE, FDE). It can be observed that, by learning from a dataset containing anomalous samples, the model effectively distinguishes anomalous data from normal data based on the distribution patterns of the normal samples, thereby ensuring a certain level of accuracy. To further validate the model's ability to identify anomalous trajectory points, we conducted an analysis using gradient information to evaluate the influence of each input trajectory point on the regression results. This allowed us to visualize the importance of input features on the model's prediction



(a) Visualization and Interpretability of Normal Sample Trajectory Predictions



(b) Visualization and Interpretability of Anomalous Sample Trajectory Predictions

Fig. 11. Interpretability analysis results for anomalous trajectory prediction.

output. In deep learning, gradients reflect the model's sensitivity to changes in the input, enabling the rapid identification of the most influential features under a given input. As shown in Fig. 11, Fig. 11(a) visualizes a normal trajectory sample (containing seven historical trajectory points and three predicted trajectory points) along with the importance of each input trajectory point to the model's prediction. It can be seen that the later trajectory points in the sequence are crucial for predicting the subsequent trajectory points. Fig. 11(b) shows the visualization of the same normal trajectory sample where the fifth historical trajectory point is replaced by an anomalous data point, and the importance of each input trajectory point to the model's prediction is also displayed. The results show that after the replacement of point 5, its importance to the output significantly decreased to near zero, while the importance of the other points slightly increased. Moreover, the prediction result was nearly identical to the ground truth, which verifies that the model effectively identified the anomalous data point and ignored it.

5.5. Comparison experiment

5.5.1. Model hyper-parameters and training

All baseline models and the proposed PosTraFormer network were executed on a server equipped with an NVIDIA GeForce 4090 GPU. All models were constructed using the PyTorch deep learning framework. The baseline models include the FairMOT (Zhang and Ge, 2022), LSTM, Gated Recurrent Unit (GRU), Bidirectional Long Short-Term Memory (BiLSTM), as well as LSTM-Attention, GRU-Attention, and BiLSTM-Attention. The hidden layer dimension for all models was set to 64, and the batch size was set to 32. The dropout layer with a dropout probability of 0.1 was employed to prevent overfitting. The ReLU activation function was used for all models. The epoch size for all methods was set to 300. The initial learning rate was set to 0.001, and the Adam optimizer was utilized, resulting in dynamic adjustment of the learning rate at each epoch based on the model's performance and loss function. All methods employed the smooth L1 loss as the loss function for model training. For the PosTraFormer, the number of encoder and decoder layers was set to 2, the number of self-attention heads was set to 2, and the dimension of the feedforward network within each layer was set to 64.

From Table 4, the overall loss of PosTraFormer is 0.0156, while that of the suboptimal model, LSTM, is 0.0387. From Fig. 12, PosTraFormer has the fastest convergence speed. An interesting finding is that the simpler LSTM model has a smoother convergence curve in modular building scenarios, which is significantly better than many deep learning methods. Although the model proposed in this paper converges faster, the oscillations are more significant in the early stage of training. This is because the transformer can process the data in parallel, unlike the LSTM which processes the data sequentially. This characteristic that processes all the inputs at once through the self-attention mechanism allows a faster training process, but at the same time it may lead to a more oscillatory loss at the initial learning stage. Because the model tries to solve the complex relationship between all inputs and outputs simultaneously at the beginning, this global initialization may

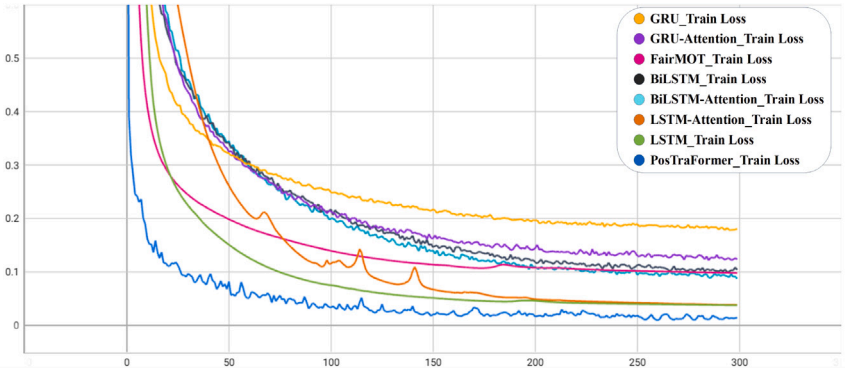


Fig. 12. Training loss of different models.

not perform as stable as the gradual optimization of LSTM. In addition, our proposed model has larger parameters and relatively high model capacity. In the early stage of training, the large capacity may make the model show greater volatility when trying various data features. In contrast, LSTM has a relatively smooth training process due to its smaller parameters.

5.5.2. Comparative experiments in different scenarios

During the actual module installation process, workers assist the crane operators in securing the hook onto the module. The crane lifts modules from the buffer zone to a certain height to avoid obstacles. Subsequently, the crane rotates to transfer the module above the target area, where the module undergoes translation and rotation. The winch mechanism then controls the gradual descent of the module. At the target position, construction workers rotate the module to the appropriate angle and assist the crane in moving the module to its final position. Consequently, the hoisting process involves different scenarios with distinct trajectories. To enhance the adaptability of the DTPTS system to various lifting processes and improve the accuracy and robustness of module trajectory prediction, we conducted real-time pose estimation and trajectory prediction tests for different scenarios. In this study, four stacking scenarios were selected: (1) Module lifting process, (2) Module rotation and translation process, (3) Module descent process, and (4) Module self-rotation and adjustment process. The schematic diagrams for the four scenarios are illustrated in Fig. 13. To fully evaluate the effectiveness and robustness of the models, visualization 3D module lifting trajectory comparison and quantitative analysis were performed separately.

(1) Visualization experiments.

Fig. 14 shows the comparison of all the models in four different scenarios. PosTraFormer shows better accuracy in all four scenarios. In the lifting scenario, PosTraFormer works best. Although LSTM-Attention can achieve accurate estimation of angles, there is a large gap in localization accuracy. LSTM is the opposite of LSTM-Attention, with a large gap in angular localization. It may be attributed to the fact that the lifting process is faster in comparison to the rotation process, and the LSTM model may not be able to capture the angle in the lifting scenario due to the difficulty of capturing the angle in the rotation process. This may be due to the fact that the lifting process is faster compared to the rotation process, and the LSTM model may have a gap with the PosTraFormer model in the lifting scene due to the difficulty in capturing the long-distance dependencies. In translational rotation, the effect of PosTraFormer is significantly improved compared to other models. While in descent, the prediction accuracy of PosTraFormer is lower than that of LSTM and LSTM-Attention, and it is possible that PosTraFormer is difficult to capture the subtle changes and critical attitude adjustments during module descent, which leads to lower prediction accuracy. During the spin adjustment stage, the PosTraFormer model may be able to understand the complexity of module spin adjustment better. Through the visualization experiments, it is well demonstrated that PosTraFormer can better capture global patterns and long range dependencies, resulting in more accurate position and attitude predictions.

(2) Quantitative analysis.

Fig. 15 shows the results of boxplots for four different scenarios using different models. In the four application scenarios, the proposed PosTraFormer performs the best, with FDE and ADE significantly lower than the other compared models and shows strong robustness. In Scenario 1 (Lifting), LSTM-Attention, BiLSTM and BiLSTM-Attention has the worst performance in the FDE metrics, and the models are less robust. LSTM performs the best among the comparison models, except for the model proposed in this paper. Compared to LSTM, PosTraFormer's attention mechanism and spatio-temporal feature fusion network can better capture the spatio-temporal relationship of the trajectory points, resulting in smaller prediction error and higher robustness. In Scenario 2 (Rotation & Translation), although the ADE metric of PosTraFormer shows outliers, the accuracy and robustness are still better than other models. This is because translations and rotations have greater pose changes and high action complexity, which increases the difficulty of prediction, as the model needs to accurately capture and predict dynamics in multiple dimensions at the same time. The LSTM performs the worst, as the rotational trajectory has higher dynamic complexity compared to a simple ascending trajectory. Rotation is designed to have continuous changes in orientation, and while LSTM is good at handling time series data, its internal structure based on gating mechanisms is not sufficient to fully capture the high frequency changes in rotational motion. Rotational motion is not only continuous in time, but also involves complex changes in space. PosTraFormer performs best in Scenario 3 (descending). LSTM-Attention outperforms LSTM as well as BiLSTM due to the inclusion of the Attention mechanism that allows the model to process each time step information with more focus on the other time steps that are most relevant to the current prediction. In Scenario 4, PosTraFormer performs much better than the other models, and in the complex scenario of self-rotation adjustment, PosTraFormer is much more sensitive in capturing the rotational spatial features as well as the temporal features.

From Table 4, it can also be observed that the ADE and FDE of PosTraFormer are the lowest in all four scenarios. Except for PosTraFormer, LSTM-Attention performs sub-optimally in scenarios 3 and 4. In Scenario 1, the method proposed in this paper outperforms GRU by 37.29% in FDE and outperforms BiLSTM-Attention by 45.6% in ADE, respectively. In Scenario 2, the PosTraFormer exhibits a deterioration in performance on FDE and ADE compared to Scenario 1, showing a 5.52% and 17.01% increase in error, respectively. In Scenario 3, the PosTraFormer performs best in the four scenarios. In scenario 4, the method proposed in this paper outperforms LSTM-Attention by 29.87% in FDE and 18.30% in ADE, respectively. From the results, the difference in performance between PosTraFormer and the suboptimal

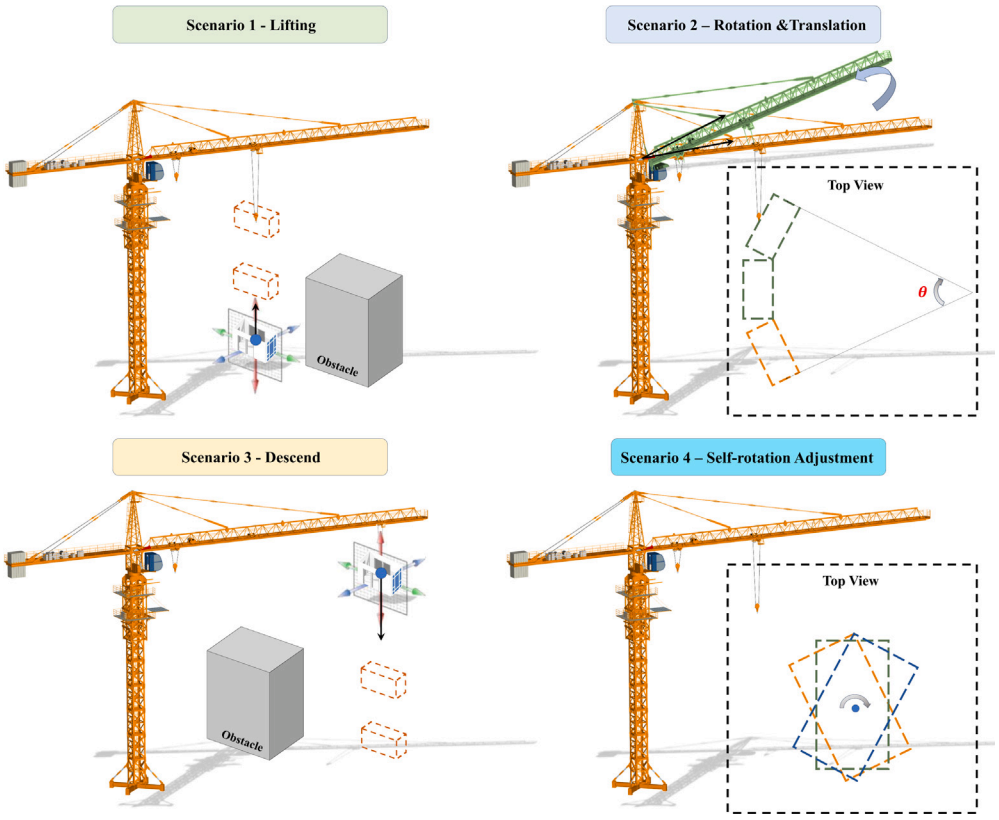


Fig. 13. The schematic diagrams for the four scenarios.

Table 4
Comparison of evaluation metrics and loss.

	Train	Scenario 1		Scenario 2		Scenario 3		Scenario 4	
Model	Loss	FDE	ADE	FDE	ADE	FDE	ADE	FDE	ADE
FairMOT	0.0989	0.1240	0.1480	0.1121	0.1476	0.1067	0.1099	0.0720	0.0814
GRU	0.1814	**0.0665**	0.0950	**0.0672**	**0.0959**	0.1121	0.1166	0.0815	0.0859
GRU-Attention	0.1245	0.1092	0.1232	0.0987	0.1167	0.1290	0.1342	0.0675	0.0835
LSTM	0.0387	0.0676	0.0965	0.0751	0.1098	0.0993	0.1216	0.0717	0.0794
LSTM-Attention	0.0395	0.2096	0.1571	0.1901	0.1705	**0.0810**	**0.0922**	**0.0616**	**0.0612**
BiLSTM	0.1045	0.1325	0.1216	0.1237	0.7155	0.1467	0.6867	0.1197	0.5367
BiLSTM-Attention	0.0885	0.1394	**0.0875**	0.1491	0.1323	0.1079	0.1233	0.0969	0.0890
Ours(PosTraFormer)	0.0156	**0.0417**	**0.0476**	**0.0440**	**0.0557**	**0.0407**	**0.0536**	**0.0432**	**0.0500**

Red text represents optimal model data.
Blue text is sub-optimal model data.

model is the smallest in the simpler lifting scenario (Scenario 1), and PosTraFormer shows much better performance than the other models in the complex rotating and translating scenario (Scenario 2), which shows that the model proposed in this paper is much more adaptive and robust to the complex architectural scenarios.

5.5.3. Complexity analysis

To assess the feasibility of deploying the PosTraFormer in actual applications, we conducted a comprehensive analysis of its computational efficiency relative to advanced methods. Our evaluation focused on critical performance metrics that are paramount in scenarios where computational resources and runtime efficiency are essential. The experimental framework encompassed various tests designed to measure key indicators such as the number of parameters (Params), floating point operations (FLOPs), and frames per second (FPS). FLOPs represent the number of floating point operations executed during task performance. FPS measures the number of frames processed per second. This metric is particularly critical for applications requiring real-time processing. In our context, given that the data involves three-dimensional position coordinates and poses, FPS specifically refers to the number of samples the model processes per second. This is quantified by measuring how many samples are processed by the model within a stipulated timeframe. Our findings, presented in Table 5, reveal that the PosTraFormer possesses a significantly larger number of parameters compared to other models evaluated. This extensive parameterization, while crucial for capturing complex spatio-temporal relationships, does impose a notable impact on the model's computational efficiency. Specifically, the increased number of parameters correlates with a reduction in the FPS (frames per second) rate, making the PosTraFormer the slowest among the tested models in terms of data processing speed. Despite this limitation, it is noteworthy that the PosTraFormer still manages to achieve an FPS of 288.8f/s. Although this is lower compared to other models, it remains sufficient for meeting the real-time processing demands of many applications, while also providing a high level of accuracy. Thus, while the large parameter count of the PosTraFormer presents certain challenges, particularly in terms of speed, its ability to fulfill real-time requirements alongside maintaining accuracy renders it a considerable option for deployment in practical settings where complex data relationships are prevalent.

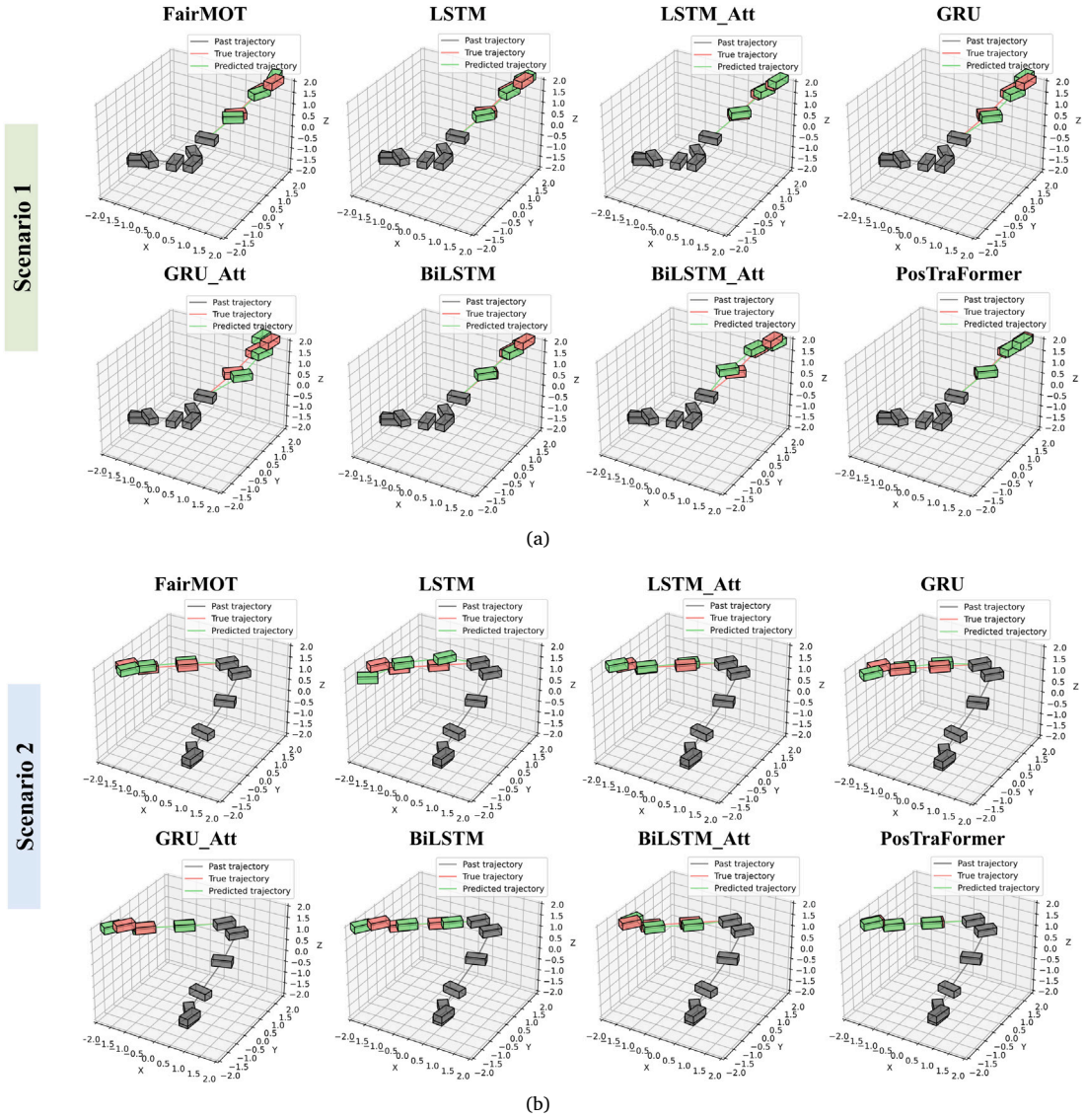


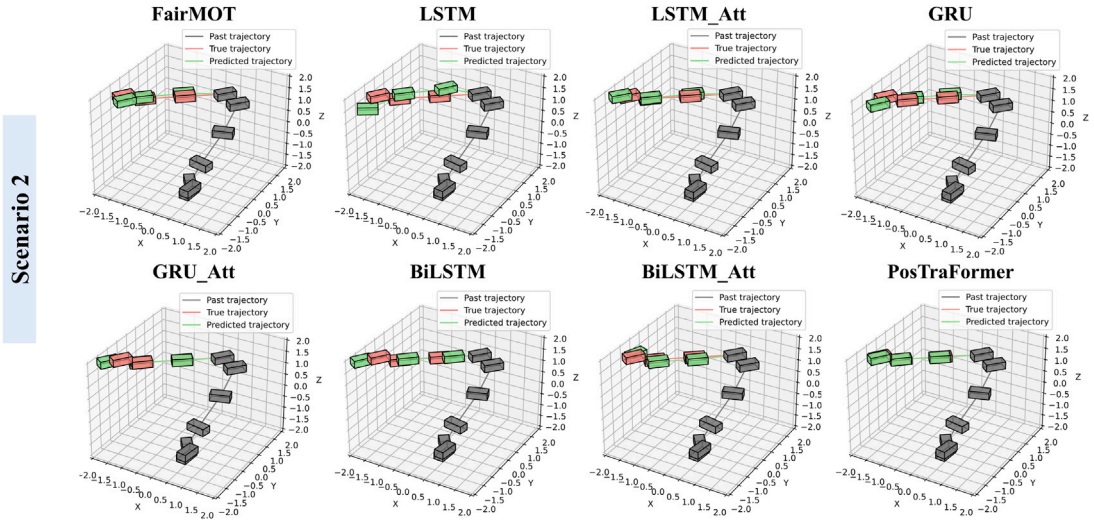
Fig. 14. 3D visualization of trajectory prediction in different scenarios. The gray rectangular block in the figure indicates the past real trajectory of the module, including the 3D position of the module as well as the rotational attitude; the red rectangular block indicates the real trajectory of the module that needs to be predicted, and the green rectangular block indicates the predicted trajectory of the module.

Table 5
Results of complexity experiments for PosTraFormer and state-of-the-art methods.

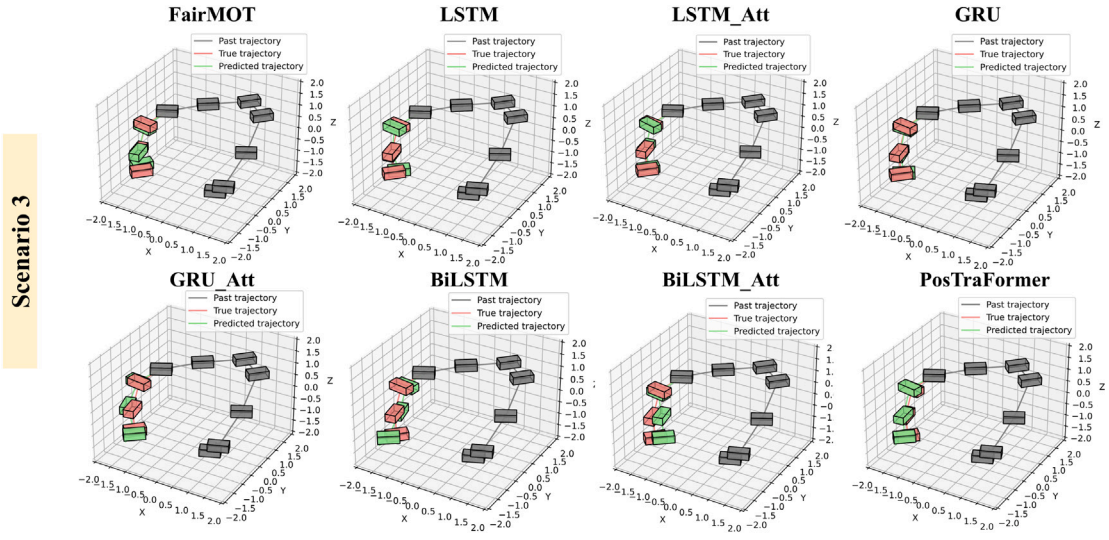
Method	Params(K)	FLOPs(K)	FPS(f/s)
GRU	14.2	87.9	1311.3
GRU-Attention	18.4	137.1	946.9
LSTM	18.9	116.5	811.6
LSTM-Attention	23.0	165.5	774.8
BiLSTM	37.8	232.7	704.5
BiLSTM-Attention	54.3	429.3	420.4
FairMOT	5399.1	32 396.3	300.9
PosTraFormer	5661.7	33 979.2	288.8

5.6. Ablation study

To verify the effectiveness of the various components of the proposed PosTraFormer model, we conducted an ablation study. This study involved comparing the performance of the full version of the model with two ablated versions: one without spatial structure modeling (W/O Spatial structure modeling) and another without temporal relationship modeling (W/O Temporal relationship modeling). The comparison was made using the FDE and ADE metrics across the four stacking scenarios. Table 6 presents the comparison results, showing that the full version of PosTraFormer achieved the best performance. Especially, the module rotation and translation process and self-rotation and adjustment process involve complex spatial movements, requiring precise spatial structure modeling to capture the module's orientation and position changes. The version without spatial structure modeling (W/O Spatial structure modeling) performed poorly here, highlighting the need for accurate spatial understanding to predict the trajectory effectively. Furthermore, the module lifting and descending process heavily depends on temporal factors, with minimal spatial complexity. The absence of temporal modeling led to higher errors, confirming that temporal relationship modeling plays a critical role in this stage. These results underscore the importance of both spatial and temporal modeling components in accurately predicting lifting trajectories and demonstrate the superiority of the complete PosTraFormer architecture.



(c)



(d)

Fig. 14. (*continued*).

Table 6
Comparison on evaluation metrics and loss.

Model	Train	Scenario 1		Scenario 2		Scenario 3		Scenario 4	
	Loss	FDE	ADE	FDE	ADE	FDE	ADE	FDE	ADE
Full version	**0.0156**	**0.0417**	**0.0476**	**0.0440**	**0.0557**	**0.0407**	**0.0536**	**0.0432**	**0.0500**
W/O Spatial modeling	0.0257	0.0544	0.0539	0.0575	0.0680	0.0511	0.0564	0.0599	0.0610
W/O Temporal modeling	0.0266	0.0604	0.0633	0.0529	0.0623	0.0590	0.0633	0.0577	0.0602

Full version: the full PosTraFormer model proposed in this paper, with spatio-temporal relationship modeling as well. W/O Spatial modeling: no spatial structure modeling. W/O Temporal modeling: no temporal relationship modeling.

6. Practical implementation of DTPTS in case study

6.1. Case scenarios and settings

Data processing. We captured 70 sets of real-time pose data of the building modules from the installed sensors. Fig. 16 illustrates a complete and authentic trajectory during the module installation process within the case data. The three-dimensional pose data corresponding to each point of the trajectory can be found in Appendix. Due to the limited amount of data, it is challenging to efficiently fit a large number of model parameters. Therefore, we fine-tuned the pretrained PosTraFormer model using the case data, as the model, pretrained on simulated data, already possesses prior knowledge about the operational patterns of the lifting trajectories. For the fine-tuning process, a learning rate of 5e-5 was chosen to allow the model to adapt gradually to the new data, minimizing the risk of catastrophic forgetting of the knowledge acquired during pretraining. The batch size was set to 4 to balance computational efficiency and the stability of gradient updates. Furthermore, we conducted fine-tuning over 10 epochs, using the Adam optimizer with a weight decay of 0.01 to prevent overfitting. Additionally, 20% of the case data was reserved for validation to monitor the model's generalization performance during

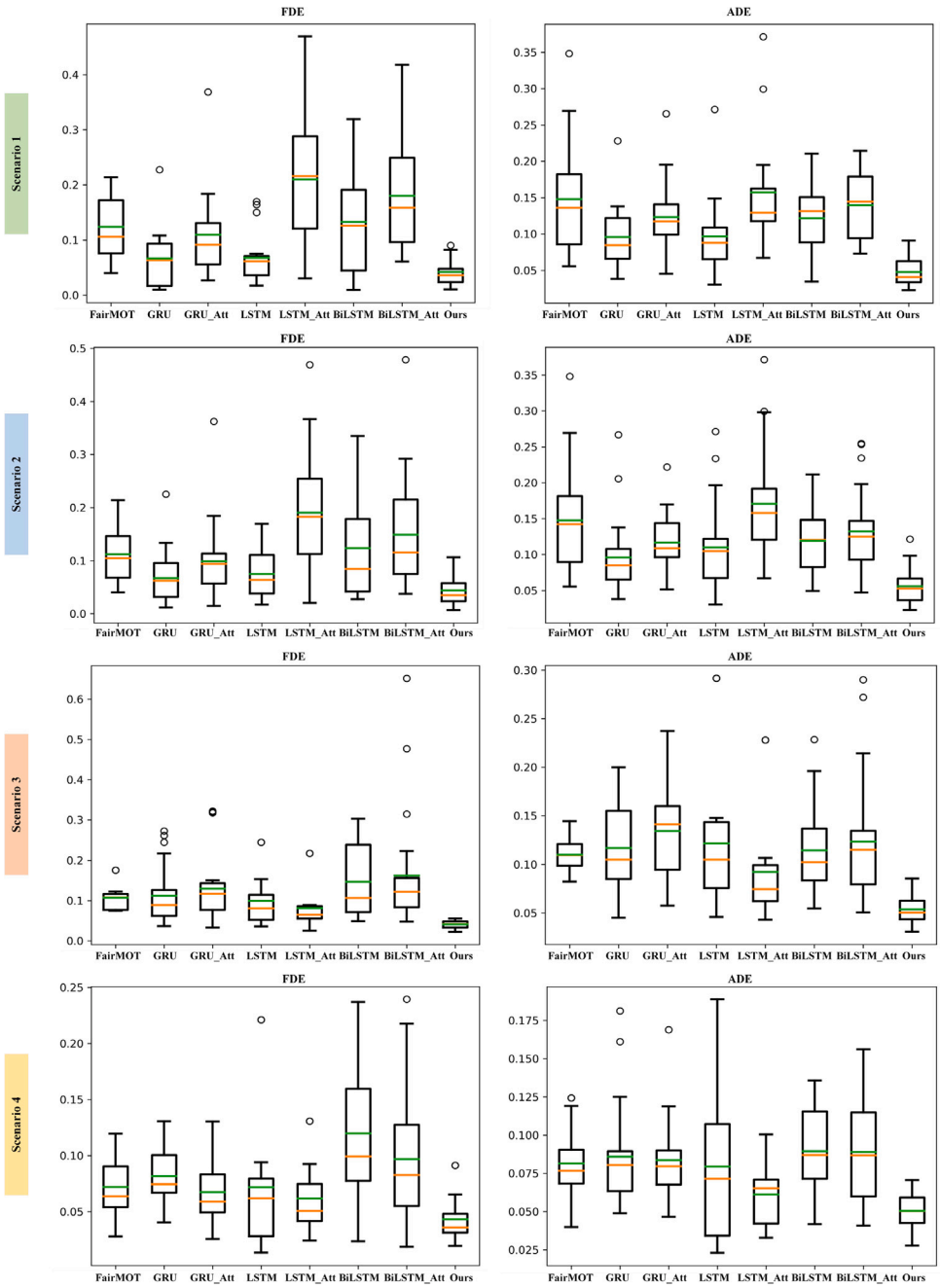


Fig. 15. The boxplots in different scenarios. The box plots with FDE & ADE metrics in the four scenarios of Lifting (Scenario 1), Rotation& Translation (Scenario 2), Descend (Scenario 3), and Self-rotation Adjustment (Scenario 4) are represented.

fine-tuning. The fine-tuned model generated predicted pose trajectories, which were continuously transmitted into DTPTS, allowing for real-time trajectory control and monitoring during the lifting process.

6.2. Modeling and visualization in the DTPTS

DTPTS provided comprehensive real-time monitoring and control of the entire operation in the lifting process. As shown in Fig. 17, the system's interface allowed for detailed oversight of the critical parameters, including the wind speed, load, and position of the modules, as well as the pose of the workers and machinery involved. The interface can offer a highly intuitive visualization, enabling the operators to seamlessly interact with the system in real time. Fig. 18 illustrates the predicted future poses of the building modules during the lifting process. Using the fine-tuned PosTraFormer model, the system could predict the modules' future landing positions, enabling the operators to control the lifting direction with increased precision. This predictive capability played a crucial role in optimizing the lifting trajectory, reducing the likelihood of misalignment, and improving overall efficiency. Furthermore, both Fig. 18 and Fig. 19 showcase a critical feature of the system: real-time collision avoidance. During the lifting process, the DT continuously projected transparent protective shields around workers, modules, and vehicles to prevent any potential collisions. These shields dynamically updated as the operation progressed, ensuring a high level of safety by providing both visual and automated collision prevention mechanisms.

By integrating real-time data from IoT sensors and predictive models, DTPTS offered an unprecedented level of control and safety in complex construction scenarios. The visualization of future module trajectories and the implementation of protective shields allowed for both

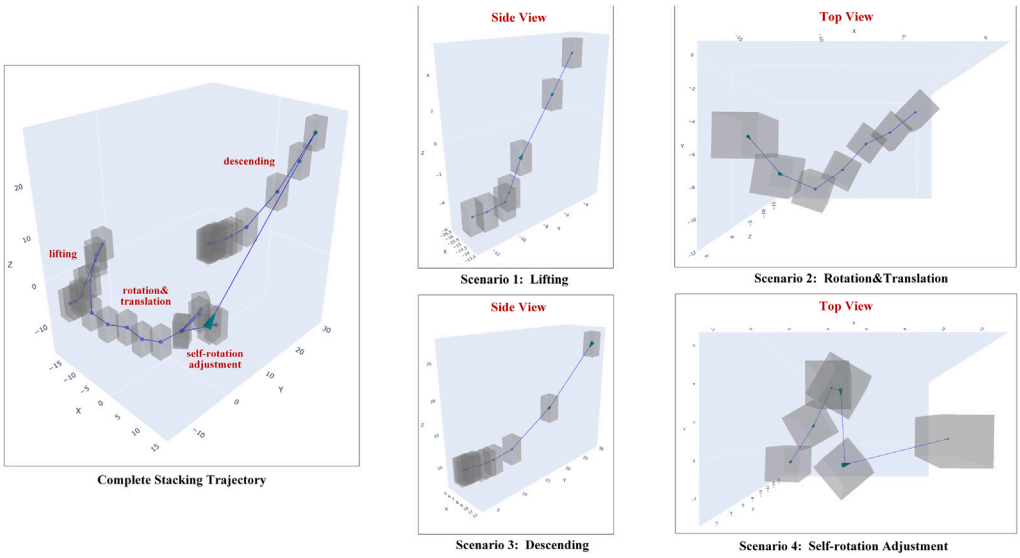


Fig. 16. Visualization of complete trajectories of modules in case studies. This trajectory encompasses four distinct phases: Lifting, Rotation & Translation, Descending, and Self-Rotation Adjustment.

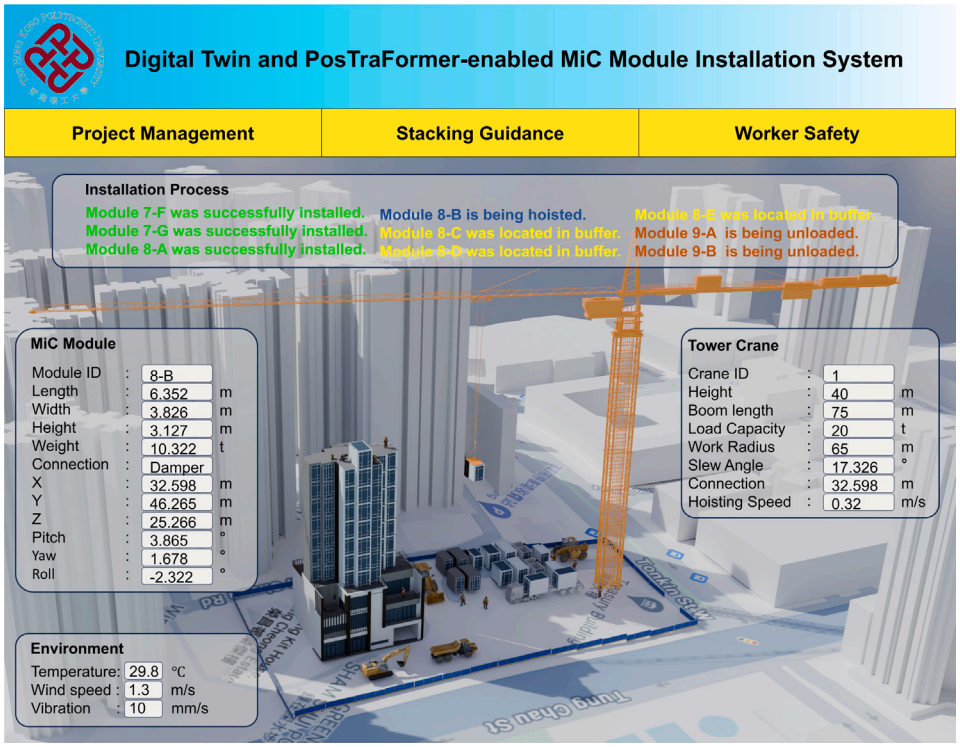


Fig. 17. Real-time monitoring interface of the lifting process of DTPTS. The green font on the page displays the installed modules and the blue font displays information about the modules being installed. The yellow font shows the location of the module to be installed. The orange font shows the status of the module that has not arrived. The two small boxes on the left side of the page show the detailed data of the module being installed and the environmental information of the current installation site. The small box on the right side of the page shows the detailed information of the tower crane that is working, including the crane ID, height, and so on.

preventive and responsive safety measures, significantly improving the reliability of the lifting operations.

6.3. Case study on the application of PosTraFormer

The performance of the proposed PosTraFormer model was rigorously evaluated by comparing the predicted module trajectories with actual data collected from the construction site. Table 7 presents the

Table 7
Results of evaluation metrics of PosTraFormer on case study.

Model	Fine-tuning loss	FDE	ADE
PosTraFormer	0.0209	0.0277	0.0319

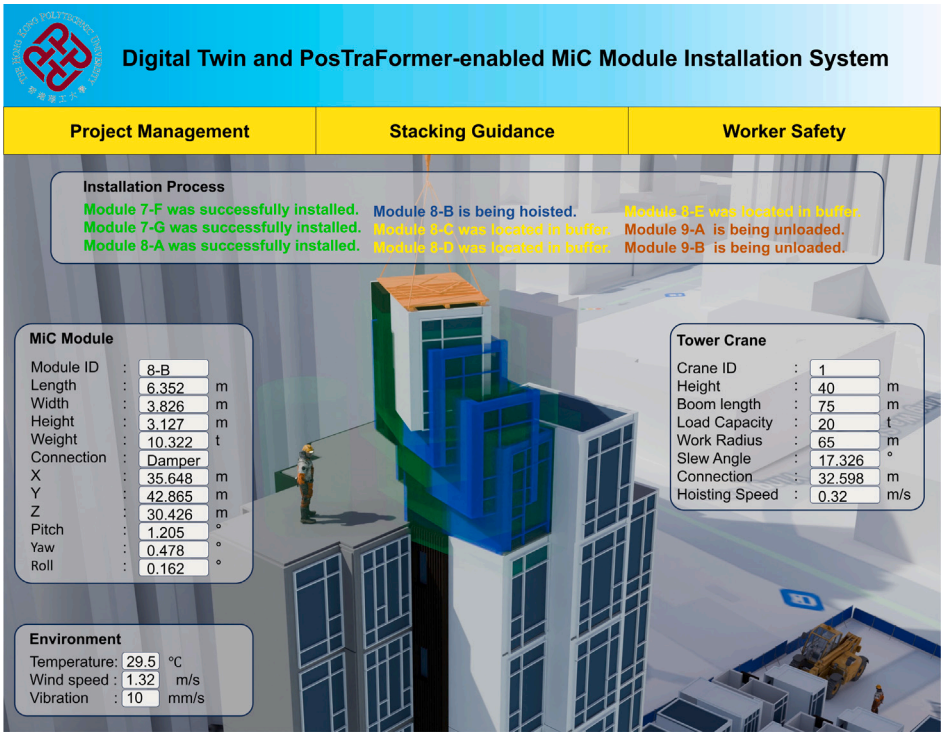


Fig. 18. Visualization of module prediction trajectories in DTPTS. The blue module in the figure indicates the predicted position and attitude information for the next three moments of the module being installed, and the green outer box indicates the safe range of the module.

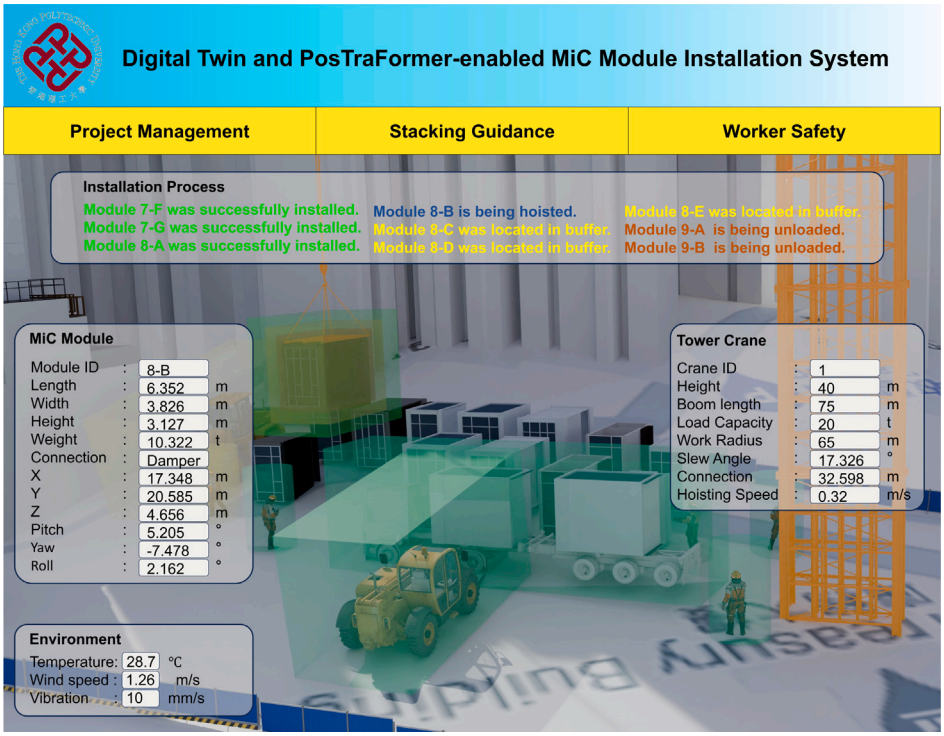


Fig. 19. Visualization of module collision detection in DTPTS. The green outer boxes for vehicles, workers, and modules in the figure indicate the security scope of the corresponding physical entities.

Table A.1
A collected trajectory data from experimental case.

Points No.	X(m)	Y(m)	Z(m)	Yaw(°)	Pitch(°)	Roll(°)
1	−15.224588	−12.163367	−4.902785	−0.824843	−0.857555	−0.863072
2	−14.536419	−11.514773	−4.467018	−0.805771	−0.857329	−0.869932
3	−14.061547	−10.480089	−3.956465	−0.797643	−0.872522	−0.893783
4	−14.309717	−9.947436	−3.607899	−0.790105	−0.877094	−0.902527
5	−14.866042	−8.502885	−1.793387	−0.740774	−0.863099	−0.896062
6	−15.472266	−5.231674	1.732744	−0.673174	−0.856000	−0.898797
7	−15.930172	−2.944987	4.259672	−0.632801	−0.857171	−0.905762
8	−15.685264	−4.828900	2.753659	−0.675698	−0.856082	−0.903528
9	−14.221419	−7.824228	−1.395628	−9.235649	−0.873617	−0.912250
10	−11.813098	−9.714396	−6.699606	−17.554236	−0.949978	−0.970355
11	−9.192876	−7.979134	−9.133282	−29.238548	−1.073219	−1.079314
12	−6.677368	−5.214999	−10.063085	−36.178953	−1.233729	−1.231491
13	−3.836690	−3.891088	−12.169618	−40.332456	−1.427672	−1.414563
14	−0.853964	−1.543582	−12.571989	−45.715864	−1.550222	−1.527266
15	1.704466	1.963309	−10.421814	33.945761	−1.490790	−1.459106
16	3.575114	4.902052	−7.186855	96.744523	−1.244931	−1.201145
17	4.406378	4.435576	−5.304950	152.245761	−0.837831	−0.778827
18	4.643599	−1.171897	−7.412717	227.841370	−0.321322	−0.248200
19	11.602638	1.022637	−3.604798	−0.364579	1.055441	1.054821
20	12.119750	−1.402003	−1.766830	1.004154	1.088280	1.076127
21	11.656340	22.935021	21.516989	1.412902	1.123750	1.058916
22	10.923767	28.438098	25.411809	1.406005	1.080675	1.021005
23	10.043885	18.861804	16.240076	1.174177	1.025938	1.003064
24	8.776831	11.801373	11.086385	1.064349	1.012625	1.008778
25	7.551312	9.112031	9.835836	1.025306	0.996163	0.997352
26	7.028228	8.210035	9.503051	1.002911	0.978639	0.981321
27	6.665509	7.477639	9.300393	1.000255	0.978303	0.981339
28	5.977748	6.400756	9.189712	1.015354	0.994093	0.996736
29	5.435445	5.305624	8.974769	1.016127	0.995969	0.998482
30	5.415329	5.095972	8.879453	0.995652	0.974023	0.976417
31	5.759804	5.568063	8.982664	0.982634	0.958929	0.961425
32	5.986436	6.073495	9.118789	0.979645	0.955596	0.958660

The sequence of labels for the trajectory points in the table represents the temporal relationship between the trajectories.

results of this case study. It can be observed that the fine-tuning of PosTraFormer on actual data resulted in a notably low fine-tuning loss, which further validates the effectiveness of pretraining on simulated data. Moreover, the fine-tuned PosTraFormer exhibited strong performance on both metrics FDE and ADE, consistently providing accurate predictions of the building modules' future positions. This high level of precision not only reduced the risk of misalignment during the lifting process but also enhanced operational safety and efficiency, making it a valuable tool for predictive trajectory control in DT systems.

7. Conclusions

In this paper, we proposed a framework for MiC onsite assembly process services based on DT. On the one hand, real-time pose estimation information of modules is obtained through UWBs and IMUs sensors. On the other hand, a novel spatial–temporal feature fusion modeling network named PosTraFormer is proposed for trajectory prediction during the module installation process. This framework aims to provide guidance and monitoring services for relevant stakeholders during the stacking process. Furthermore, due to the scarcity of trajectory datasets in existing literature for module stacking processes, we collected data based on physical simulation models, constructing a dataset of module pose and trajectory during the installation process, and enhancing it through sliding window techniques. Through the comparative experiments and case studies, it can be observed that our proposed PosTraFormer achieves higher trajectory prediction performance. This renders it suitable for construction environments, enhancing accuracy and minimizing collision risks during the module stacking process. Additionally, real-time pose estimation based on IMUs and UWBs can reflect the position and orientation of modules in building scenarios, enabling construction site workers and managers to monitor building dynamics in real-time, and facilitating operational management.

CRediT authorship contribution statement

Yujie Han: Writing – original draft, Methodology, Data curation. **Kexin Sun:** Validation, Software. **Zhiheng Zhao:** Writing – review & editing. **Yantao Yu:** Formal analysis. **George Q. Huang:** Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix. Additional data

In the appendix, we present a trajectory data sample collected from the experimental case in Table A.1. This data includes the coordinates (X, Y, Z) and angle information (yaw, pitch, roll) for 32 trajectory points, providing detailed insight into the entire motion process, including position and attitude changes.

Data availability

Data will be made available on request.

References

Akhavian, R., Amani, M., Mootz, J., Ashe, R., Beheshti, B., 2025. Building information models to robot-ready site digital twins (BIM2rdt): An agentic AI safety-first framework. arXiv preprint arXiv:2509.20705.

Alduais, M., Li, X., Mei, Q., 2025. Trajectory forecasting for worker safety in construction using transformer and graph attention networks. In: Proceedings of the Sixth International Confer, vol. 22, pp. 781–791.

Ali, A., Ullah, I., Ahmad, S., Wu, Z., Li, J., Bai, X., 2025. An attention-driven spatio-temporal deep hybrid neural networks for traffic flow prediction in transportation systems. IEEE Trans. Intell. Transp. Syst. 26 (9), 14154–14168. http://dx.doi.org/10.1109/TITS.2025.3540852.

Azar, E.R., Feng, C., Kamat, V.R., 2015. Feasibility of in-plane articulation monitoring of excavator arm using planar marker tracking. J. Inf. Technol. Constr. (ITcon) 20 (15), 213–229, URL: https://api.semanticscholar.org/CorpusID:14551776.

Brachmann, E., Michel, F., Krull, A., Yang, M.Y., Gumhold, S., Rother, C., 2016. Uncertainty-driven 6D pose estimation of objects and scenes from a single RGB image. In: 2016 IEEE Conference on Computer Vision and Pattern Recognition. CVPR, pp. 3364–3372. http://dx.doi.org/10.1109/CVPR.2016.366, URL: https://api.semanticscholar.org/CorpusID:16805301.

Cheng, M.-Y., Chiu, K.-C., Hsieh, Y.-M., Yang, I.-T., Chou, J.-S., Wu, Y.-W., 2017. BIM integrated smart monitoring technique for building fire prevention and disaster relief. Autom. Constr. (ISSN: 0926-5805) 84, 14–30. http://dx.doi.org/10.1016/j.autcon.2017.08.027, URL: https://www.sciencedirect.com/science/article/pii/S0926580516304149.

Cheng, Y., Lin, F., Wang, W., Zhang, J., 2022. Vision-based trajectory monitoring for assembly alignment of precast concrete bridge components. Autom. Constr. 140, 104350. http://dx.doi.org/10.1016/j.autcon.2022.104350, URL: https://www.sciencedirect.com/science/article/pii/S0926580522002230.

Cheng, H., Stock, J., Xhonneux, A., Çakmak, H.K., Hagenmeyer, V., 2025. Construction and control of validated highly configurable multi-physics building models for the sustainability analysis of multi-energy systems in a co-simulation setup. In: SoutheastCon 2025. IEEE, pp. 19–28.

Cheng, Y., Zhu, H., Sun, Y., Acar, C., Jing, W., Wu, Y., Li, L., Tan, C., Lim, J.-H., 2021. 6D pose estimation with correlation fusion. In: 2020 25th International Conference on Pattern Recognition. ICPR, pp. 2988–2994. http://dx.doi.org/10.1109/ICPR48806.2021.9412238, URL: https://api.semanticscholar.org/CorpusID:203593673.

Fu, Y., Lu, W., Dong, Z., Fang, Y., 2025. A synthetic data-enhanced method for automated 3D pose recognition of construction workers. Expert Syst. Appl. 128768.

Gao, Y., Yang, K., Yue, Y., Wu, Y., 2025. A vehicle trajectory prediction model that integrates spatial interaction and multiscale temporal features. Sci. Rep. 15 (1), 8217.

Groves, P.D., 2011. Shadow matching: A new GNSS positioning technique for urban canyons. J. Navig. 64 (3), 417–430. http://dx.doi.org/10.1017/S0373463311000087, URL: https://api.semanticscholar.org/CorpusID:18769611.

Han, S., Lee, S., 2013. A vision-based motion capture and recognition framework for behavior-based safety management. Autom. Constr. (ISSN: 0926-5805) 35, 131–141. http://dx.doi.org/10.1016/j.autcon.2013.05.001, URL: https://www.sciencedirect.com/science/article/pii/S0926580513000514.

Han, S., Lee, S., Peña-Mora, F., 2013. Vision-based detection of unsafe actions of a construction worker: Case study of ladder climbing. J. Comput. Civ. Eng. 27 (6), 635–644. http://dx.doi.org/10.1061/(ASCE)CP.1943-5487.0000279, URL: https://api.semanticscholar.org/CorpusID:111178627.

Han, Z., Sun, H., Huang, J., Xu, J., Tang, Y., Liu, X., 2024. Path planning algorithms for smart parking: Review and prospects. World Electr. Veh. J. 15 (7), 322.

Hasan, S.M., Lee, K., Moon, D., Kwon, S., Jinwoo, S., Lee, S., 2022. Augmented reality and digital twin system for interaction with construction machinery. J. Asian Archit. Build. Eng. 21 (2), 564–574. http://dx.doi.org/10.1080/13467581.2020.1869557, URL: https://doi.org/10.1080/13467581.2020.1869557.

Hosamo, H.H., Svennevig, P.R., Svidt, K., Han, D., Nielsen, H.K., 2022. A digital twin predictive maintenance framework of air handling units based on automatic fault detection and diagnostics. Energy Build. (ISSN: 0378-7788) 261, 111988. http://dx.doi.org/10.1016/j.enbuild.2022.111988, URL: https://www.sciencedirect.com/science/article/pii/S0378778822001591.

Hu, D., Li, S., Cai, J., Hu, Y., 2020. Toward intelligent workplace: Prediction-enabled proactive planning for human-robot coexistence on unstructured construction sites. In: 2020 Winter Simulation Conference. WSC, pp. 2412–2423. http://dx.doi.org/10.1109/WSC48552.2020.9384077, URL: https://api.semanticscholar.org/CorpusID:230125231.

Jiang, W., Ding, L., Zhou, C., 2020. Cyber physical system for safety management in smart construction site. Eng. Constr. Archit. Manag. ahead-of-print, http://dx.doi.org/10.1108/ECAM-10-2019-0578, URL: https://api.semanticscholar.org/CorpusID:219002774.

Jiang, Y., Li, M., Guo, D., Wu, W., Zhong, R.Y., Huang, G.Q., 2022a. Digital twin-enabled smart modular integrated construction system for on-site assembly. Comput. Ind. (ISSN: 0166-3615) 136, 103594. http://dx.doi.org/10.1016/j.compind.2021.103594, URL: https://www.sciencedirect.com/science/article/pii/S0166361521002013.

Jiang, Y., Li, M., Li, M., Liu, X., Zhong, R.Y., Pan, W., Huang, G.Q., 2022b. Digital twin-enabled real-time synchronization for planning, scheduling, and execution in precast on-site assembly. Autom. Constr. (ISSN: 0926-5805) 141, 104397. http://dx.doi.org/10.1016/j.autcon.2022.104397, URL: https://www.sciencedirect.com/science/article/pii/S0926580522002709.

Kim, J., Chi, S., 2021. A few-shot learning approach for database-free vision-based monitoring on construction sites. Autom. Constr. (ISSN: 0926-5805) 124, 103566. http://dx.doi.org/10.1016/j.autcon.2021.103566, URL: https://www.sciencedirect.com/science/article/pii/S0926580521000170.

Kok, M., Hol, J.D., Schön, T.B., 2017. Using inertial sensors for position and orientation estimation. http://dx.doi.org/10.1561/2000000094, URL: http://ieeexplore.ieee.org/document/8187588.

Kopsida, M., Brilakis, I., 2020. Real-time volume-to-plane comparison for mixed reality–based progress monitoring. J. Comput. Civ. Eng. 34 (4), 04020016. http://dx.doi.org/10.1061/(ASCE)CP.1943-5487.0000896, URL: https://ascelibrary.org/doi/abs/10.1061/%28ASCE%29CP.1943-5487.0000896.

Kosse, S., Hagedorn, P., König, M., 2025. Semantic digital twins in construction: Developing a modular system reference architecture based on information containers. Adv. Eng. Informatics 67, 103483.

Lee, D., Lee, S., 2021. Digital twin for supply chain coordination in modular construction. Appl. Sci. (ISSN: 2076-3417) 11 (13), http://dx.doi.org/10.3390/app11135909, URL: https://www.mdpi.com/2076-3417/11/13/5909.

Lee, D., Lee, S.H., Masoud, N., Krishnan, M., Li, V.C., 2021. Integrated digital twin and blockchain framework to support accountable information sharing in construction projects. Autom. Constr. (ISSN: 0926-5805) 127, 103688. http://dx.doi.org/10.1016/j.autcon.2021.103688, URL: https://www.sciencedirect.com/science/article/pii/S0926580521001394.

Li, K., Gan, V.J., Li, M., Gao, M.Y., Tiong, R.L., Yang, Y., 2024. Automated generative design and prefabrication of precast buildings using integrated BIM and graph convolutional neural network. Dev. the Built Environ. (ISSN: 2666-1659) 18, 100418. http://dx.doi.org/10.1016/j.dibe.2024.100418, URL: https://www.sciencedirect.com/science/article/pii/S2666165924000991.

Li, N., Ma, Z., Yu, R., Li, L., 2025. LSDTs: LLM-augmented semantic digital twins for adaptive knowledge-intensive infrastructure planning. arXiv preprint arXiv:2508.06799.

Liang, C.-J., Lundeen, K.M., McGee, W., Menassa, C.C., Lee, S., Kamat, V.R., 2019. A vision-based marker-less pose estimation system for articulated construction robots. Autom. Constr. (ISSN: 0926-5805) 104, 80–94. http://dx.doi.org/10.1016/j.autcon.2019.04.004, URL: https://www.sciencedirect.com/science/article/pii/S0926580518311579.

Liu, Q., Ma, Y., Chen, L., Pedrycz, W., Skibniewski, M.J., Chen, Z.-S., 2024. Artificial intelligence for production, operations and logistics management in modular construction industry: A systematic literature review. Inf. Fusion (ISSN: 1566-2535) 109, 102423. http://dx.doi.org/10.1016/j.inffus.2024.102423, URL: https://www.sciencedirect.com/science/article/pii/S156625352400201X.

Liu, Z., Meng, X., Xing, Z., Jiang, A., 2021. Digital twin-based safety risk coupling of prefabricated building hoisting. Sensors (ISSN: 1424-8220) 21 (11), http://dx.doi.org/10.3390/s21113583, URL: https://www.mdpi.com/1424-8220/21/11/3583.

Liu, H., Ota, H., Wei, X., Hirao, Y., Perusquia-Hernandez, M., Uchiyama, H., Kiyokawa, K., 2025. Umotion: Uncertainty-driven human motion estimation from inertial and ultra-wideband units. In: Proceedings of the Computer Vision and Pattern Recognition Conference. pp. 7085–7094.

Liu, J., Qiu, Z., Wang, L., Liu, P., Cheng, G., Chen, Y., 2024. Intelligent floor plan design of modular high-rise residential building based on graph-constrained generative adversarial networks. Autom. Constr. (ISSN: 0926-5805) 159, 105264. http://dx.doi.org/10.1016/j.autcon.2023.105264, URL: https://www.sciencedirect.com/science/article/pii/S0926580523005241.

Liu, X., Xu, F., Zhang, Z., Sun, K., 2023. Fall-portent detection for construction sites based on computer vision and machine learning. Eng. Constr. Archit. Manag. http://dx.doi.org/10.1108/ecam-05-2023-0458, URL: https://api.semanticscholar.org/CorpusID:263916502.

Lu, Q., Xie, X., Parlikad, A.K., Schooling, J.M., 2020. Digital twin-enabled anomaly detection for built asset monitoring in operation and maintenance. Autom. Constr. (ISSN: 0926-5805) 118, 103277. http://dx.doi.org/10.1016/j.autcon.2020.103277, URL: https://www.sciencedirect.com/science/article/pii/S0926580520303654.

Lundeen, K.M., Dong, S., Fredricks, N., Akula, M., Seo, J., Kamat, V.R., 2016. Optical marker-based end effector pose estimation for articulated excavators. Autom. Constr. (ISSN: 0926-5805) 65, 51–64. http://dx.doi.org/10.1016/j.autcon.2016.02.003, URL: https://www.sciencedirect.com/science/article/pii/S0926580516300255.

Nguyen, T.D.H.N., Ly, D.H., Jang, H., Dinh, H.N.N., Ahn, Y., 2025. Digital twin framework to enhance facility management for relocatable modular buildings. Autom. Constr. 176, 106249.

Omar, T., Nehdi, M.L., 2016. Data acquisition technologies for construction progress tracking. Autom. Constr. (ISSN: 0926-5805) 70, 143–155. http://dx.doi.org/10.1016/j.autcon.2016.06.016, URL: https://www.sciencedirect.com/science/article/pii/S0926580516301376.

Pan, W., Hon, C.K., 2020. Briefing: Modular integrated construction for high-rise buildings. 173, (2), pp. 64–68. http://dx.doi.org/10.1680/jmuen.18.00028, URL: https://doi.org/10.1680/jmuen.18.00028,

Pan, Y., Zhang, L., 2021. A BIM-data mining integrated digital twin framework for advanced project management. Autom. Constr. (ISSN: 0926-5805) 124, 103564. http://dx.doi.org/10.1016/j.autcon.2021.103564, URL: https://www.sciencedirect.com/science/article/pii/S0926580521000157.

Partsinevelos, P., Petrakis, G., Antonopoulos, A., Fotakis, T., Bikos, S., Charokopos, Z., Tripolitsiotis, A., 2024. Aerial geodetic total station platform for precise active positioning in GNSS-degraded environments. Autom. Constr. (ISSN: 0926-5805) 158, 105216. http://dx.doi.org/10.1016/j.autcon.2023.105216, URL: https://www.sciencedirect.com/science/article/pii/S0926580523004764.

Peng, Y., Yang, X., Wei, S., Gao, X., Li, W., Wen, J.Z., 2022. 6D hybrid pose estimation in cluttered industrial scenes for robotic grasping. In: 2022 International Conference on Industrial Automation, Robotics and Control Engineering. IARCE, pp. 19–23. http://dx.doi.org/10.1109/IARCE57187.2022.00014, URL: https://api.semanticscholar.org/CorpusID:256944560.

Pregnolato, M., Gunner, S., Voyagaki, E., De Risi, R., Carhart, N., Gavriel, G., Tully, P., Tryfonas, T., Macdonald, J., Taylor, C., 2022. Towards civil engineering 4.0: Concept, workflow and application of digital twins for existing infrastructure. Autom. Constr. (ISSN: 0926-5805) 141, 104421. http://dx.doi.org/10.1016/j.autcon.2022.104421, URL: https://www.sciencedirect.com/science/article/pii/S0926580522002941.

Raffin, A., Deutschmann, B., Stulp, F., 2021. Fault-tolerant six-dof pose estimation for tendon-driven continuum mechanisms. Front. Robot. AI (ISSN: 2296-9144) 8, http://dx.doi.org/10.3389/frobt.2021.619238, URL: https://www.frontiersin.org/articles/10.3389/frobt.2021.619238.

Rangasamy, V., Yang, J.-B., 2024. The convergence of BIM, AI and IoT: Reshaping the future of prefabricated construction. J. Build. Eng. (ISSN: 2352-7102) 84, 108606. http://dx.doi.org/10.1016/j.jobe.2024.108606, URL: https://www.sciencedirect.com/science/article/pii/S2352710224001748.

Salama, T., Salah, A., Moselhi, O., 2017. Integration of offsite and onsite schedules in modular construction. In: ISARC. Proceedings of the International Symposium on Automation and Robotics in Construction. 34, IAARC Publications, http://dx.doi.org/10.22260/ISARC2017/0107, URL: https://api.semanticscholar.org/CorpusID:56221587.

Shan, S., Looi, D., Cai, Y., Ma, P., Chen, M.-T., Su, R., Young, B., Pan, W., 2019. Engineering modular integrated construction for high-rise building: a case study in Hong Kong. 172, (6), pp. 51–57. http://dx.doi.org/10.1680/jcien.18.00052, URL: https://doi.org/10.1680/jcien.18.00052,

Shang, W., Liu, X., Wang, X., Wang, X., 2023. Fatigue life prediction for SiC/al materials based on path planning algorithm considering residual stress. Chin. J. Mech. Eng. 36 (1), 24.

Song, J., Hu, S., Yang, Y., 2018. IEKF based object pose estimation for augmented reality. In: Proceedings of the 4th International Conference on Virtual Reality. In: ICVR 2018, Association for Computing Machinery, New York, NY, USA, pp. 15–20. http://dx.doi.org/10.1145/3198910.3198914, URL: https://doi.org/10.1145/3198910.3198914.

Tagliabue, L.C., Re Cecconi, F., Rinaldi, S., Ciribini, A.L.C., 2021. Data driven indoor air quality prediction in educational facilities based on IoT network. Energy Build. (ISSN: 0378-7788) 236, 110782. http://dx.doi.org/10.1016/j.enbuild.2021.110782, URL: https://www.sciencedirect.com/science/article/pii/S0378778821000669.

Tan, Y., Chen, P., Shou, W., Sadick, A.-M., 2022. Digital twin-driven approach to improving energy efficiency of indoor lighting based on computer vision and dynamic BIM. Energy Build. (ISSN: 0378-7788) 270, 112271. http://dx.doi.org/10.1016/j.enbuild.2022.112271, URL: https://www.sciencedirect.com/science/article/pii/S037877882200442X.

Tan, Z., Tu, M., He, X., Chen, B., Zhao, Y., Shang, J., 2025. A digital twin case study on prefabricated component factory for offsite manufacturing of buildings. In: Proceedings of the 2025 6th International Conference on Computer Information and Big Data Applications. pp. 416–421.

Tang, J., Luo, H., Chen, W., Wong, P.K.-Y., Cheng, J.C., 2022. IMU-based full-body pose estimation for construction machines using kinematics modeling. Autom. Constr. (ISSN: 0926-5805) 138, 104217. http://dx.doi.org/10.1016/j.autcon.2022.104217, URL: https://www.sciencedirect.com/science/article/pii/S0926580522000905.

Tang, J., Wang, M., Luo, H., Wong, P.K.-Y., Zhang, X., Chen, W., Cheng, J.C., 2023. Full-body pose estimation for excavators based on data fusion of multiple onboard sensors. Autom. Constr. (ISSN: 0926-5805) 147, 104694. http://dx.doi.org/10.1016/j.autcon.2022.104694, URL: https://www.sciencedirect.com/science/article/pii/S0926580522005647.

Tao, F., Zhang, M., 2017. Digital twin shop-floor: A new shop-floor paradigm towards smart manufacturing. IEEE Access 5, 20418–20427. http://dx.doi.org/10.1109/ACCESS.2017.2756069, URL: https://api.semanticscholar.org/CorpusID:8600183.

Tian, Z., Yu, Y., Xu, F., Zhang, Z., 2023. Dynamic hazardous proximity zone design for excavator based on 3D mechanical arm pose estimation via computer vision. J. Constr. Eng. Manag. 149 (7), 04023048. http://dx.doi.org/10.1061/JCEMD4.COENG-13071, URL: https://ascelibrary.org/doi/abs/10.1061/JCEMD4.COENG-13071.

Tuhaise, V.V., Tah, J.H.M., Abanda, F.H., 2023. Technologies for digital twin applications in construction. Autom. Constr. (ISSN: 0926-5805) 152, 104931. http://dx.doi.org/10.1016/j.autcon.2023.104931, URL: https://www.sciencedirect.com/science/article/pii/S0926580523001917.

Wang, S., Cao, J., Yu, P.S., 2022. Deep learning for spatio-temporal data mining: A survey. IEEE Trans. Knowl. Data Eng. 34 (8), 3681–3700. http://dx.doi.org/10.1109/TKDE.2020.3025580.

Wang, X., Xu, L., Zhou, L., Liu, Y., Xiong, N., Li, K.-C., 2025. Large language model-driven probabilistic trajectory prediction in the internet of things using spatio-temporal encoding and normalizing flows. Digit. Commun. Networks (ISSN: 2352-8648).

Xie, L., Misra, S., Suresh, N., Soza-Soto, J., Furuhata, T., Shimada, K., 2025. Learning 3D human–object interaction graphs from transferable context knowledge for construction monitoring. Comput. Ind. 164, 104171.

Xue, H., Huynh, D.Q., Reynolds, M., 2018. SS-LSTM: A hierarchical LSTM model for pedestrian trajectory prediction. In: 2018 IEEE Winter Conference on Applications of Computer Vision. WACV, pp. 1186–1194. http://dx.doi.org/10.1109/WACV.2018.00135, URL: https://api.semanticscholar.org/CorpusID:13663350.

Yang, Q., Mei, Q., Fan, C., Ma, M., Li, X., 2023. Environment-aware worker trajectory prediction using surveillance camera in modular construction facilities. Buildings (ISSN: 2075-5309) 13 (6), http://dx.doi.org/10.3390/buildings13061502, URL: https://www.mdpi.com/2075-5309/13/6/1502.

Yu, Y., Umer, W., Yang, X., Antwi-Afari, M.F., 2021. Posture-related data collection methods for construction workers: A review. Autom. Constr. (ISSN: 0926-5805) 124, 103538. http://dx.doi.org/10.1016/j.autcon.2020.103538, URL: https://www.sciencedirect.com/science/article/pii/S0926580520311183.

Yuan, X., Anumba, C.J., Parfitt, M.K., 2016. Cyber-physical systems for temporary structure monitoring. Autom. Constr. (ISSN: 0926-5805) 66, 1–14. http://dx.doi.org/10.1016/j.autcon.2016.02.005, URL: https://www.sciencedirect.com/science/article/pii/S0926580516300279.

Yue Zhai, Z.L., Huang, G., 2017. Production lead-time hedging and coordination in prefabricated construction supply chain management. Int. J. Prod. Res. 55 (14), 3984–4002. http://dx.doi.org/10.1080/00207543.2016.1231432, URL: https://doi.org/10.1080/00207543.2016.1231432.

Zhai, Y., Chen, K., Zhou, J.X., Cao, J., Lyu, Z., Jin, X., Shen, G.Q., Lu, W., Huang, G.Q., 2019. An internet of things-enabled BIM platform for modular integrated construction: A case study in Hong Kong. Adv. Eng. Informatics (ISSN: 1474-0346) 42, 100997. http://dx.doi.org/10.1016/j.aei.2019.100997, URL: https://www.sciencedirect.com/science/article/pii/S1474034619305701.

Zhang, M., Ge, S., 2022. Vision and trajectory–based dynamic collision prewarning mechanism for tower cranes. J. Constr. Eng. Manag. 148 (7), 04022057. http://dx.doi.org/10.1061/(ASCE)CO.1943-7862.0002309, URL: https://ascelibrary.org/doi/abs/10.1061/%28ASCE%29CO.1943-7862.0002309.

Zhang, K., Tong, S., Shi, H., 2019. Trajectory prediction of assembly alignment of columnar precast concrete members with deep learning. Symmetry (ISSN: 2073-8994) 11 (5), http://dx.doi.org/10.3390/sym11050629, URL: https://www.mdpi.com/2073-8994/11/5/629.

Zhang, X., Wang, X., Zhang, W., Wang, Y., Liu, X., Wei, D., 2024. Multi-attention network for pedestrian intention prediction based on spatio-temporal feature fusion. Proc. Inst. Mech. Eng. Part D: J. Automob. Eng. 238 (13), 4202–4215. http://dx.doi.org/10.1177/09544070231190522, URL: https://doi.org/10.1177/09544070231190522.

Zhao, Y., Cao, C., Liu, Z., 2022. A framework for prefabricated component hoisting management systems based on digital twin technology. Buildings (ISSN: 2075-5309) 12 (3), http://dx.doi.org/10.3390/buildings12030276, URL: https://www.mdpi.com/2075-5309/12/3/276.

Zheng, Z., Zhang, Z., Pan, W., 2020. Virtual prototyping- and transfer learning-enabled module detection for modular integrated construction. Autom. Constr. (ISSN: 0926-5805) 120, 103387. http://dx.doi.org/10.1016/j.autcon.2020.103387, URL: https://www.sciencedirect.com/science/article/pii/S0926580520309675.

Zhong, R.Y., Peng, Y., Xue, F., Fang, J., Zou, W., Luo, H., Thomas Ng, S., Lu, W., Shen, G.Q., Huang, G.Q., 2017. Prefabricated construction enabled by the internet-of-things. Autom. Constr. (ISSN: 0926-5805) 76, 59–70. http://dx.doi.org/10.1016/j.autcon.2017.01.006, URL: https://www.sciencedirect.com/science/article/pii/S0926580517300225.

Zhou, J., 2025. Multi-dimensional model and interactive simulation of intelligent construction based on digital twins. Sci. Rep. 15 (1), 32189.

Zhou, C., Luo, H., Fang, W., Wei, R., Ding, L., 2019. Cyber-physical-system-based safety monitoring for blind hoisting with the internet of things: A case study. Autom. Constr. (ISSN: 0926-5805) 97, 138–150. http://dx.doi.org/10.1016/j.autcon.2018.10.017, URL: https://www.sciencedirect.com/science/article/pii/S0926580518303443.

Zhu, A., Zhang, Z., Pan, W., 2022. Crane-lift path planning for high-rise modular integrated construction through metaheuristic optimization and virtual prototyping. Autom. Constr. (ISSN: 0926-5805) 141, 104434. http://dx.doi.org/10.1016/j.autcon.2022.104434, URL: https://www.sciencedirect.com/science/article/pii/S0926580522003077.

Żywanowski, K., Łysakowski, M., Nowicki, M.R., Jacques, J.T., Tadeja, S.K., Bohné, T., Skrzypczyński, P., 2025. Vision-based hand pose estimation methods for augmented reality in industry: Crowdsourced evaluation on HoloLens 2. Comput. Ind. 171, 104328.